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Doda et al.

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(54) **HEATER INCLUDING A PLURALITY OF HEAT GENERATION MEMBERS, FIXING APPARATUS, AND IMAGE FORMING APPARATUS**

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(63) Continuation of application No. 17/883,799, filed on Aug. 9, 2022, now Pat. No. 11,774,891, which is a (Continued)

(30) **Foreign Application Priority Data**

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G03G 15/20 (2006.01)

(52) **U.S. Cl.**
CPC **G03G 15/2053** (2013.01); **G03G 15/2042** (2013.01); **G03G 15/2064** (2013.01); **G03G 2215/2035** (2013.01)

(58) **Field of Classification Search**

CPC G03G 15/2053
See application file for complete search history.

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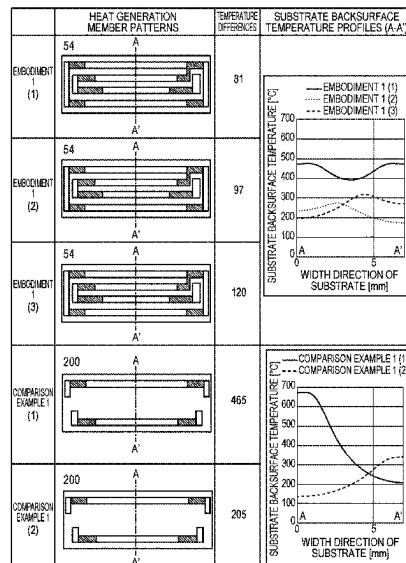
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(57) **ABSTRACT**

The heater including a substrate, a first heat generation member, a second heat generation member having a length substantially the same in a longitudinal direction as a length of the first heat generation member, a third heat generation member having a length shorter than lengths of the first heat generation member and the second heat generation member in the longitudinal direction, and a fourth heat generation member having a length shorter than length of the third heat generation member in the longitudinal direction, wherein the first heat generation member, the second heat generation member, the third heat generation member and the fourth heat generation member are arranged on the substrate.

18 Claims, 13 Drawing Sheets



Related U.S. Application Data

continuation of application No. 17/352,770, filed on Jun. 21, 2021, now Pat. No. 11,442,385, which is a continuation of application No. 16/744,669, filed on Jan. 16, 2020, now Pat. No. 11,073,778.

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FIG. 1

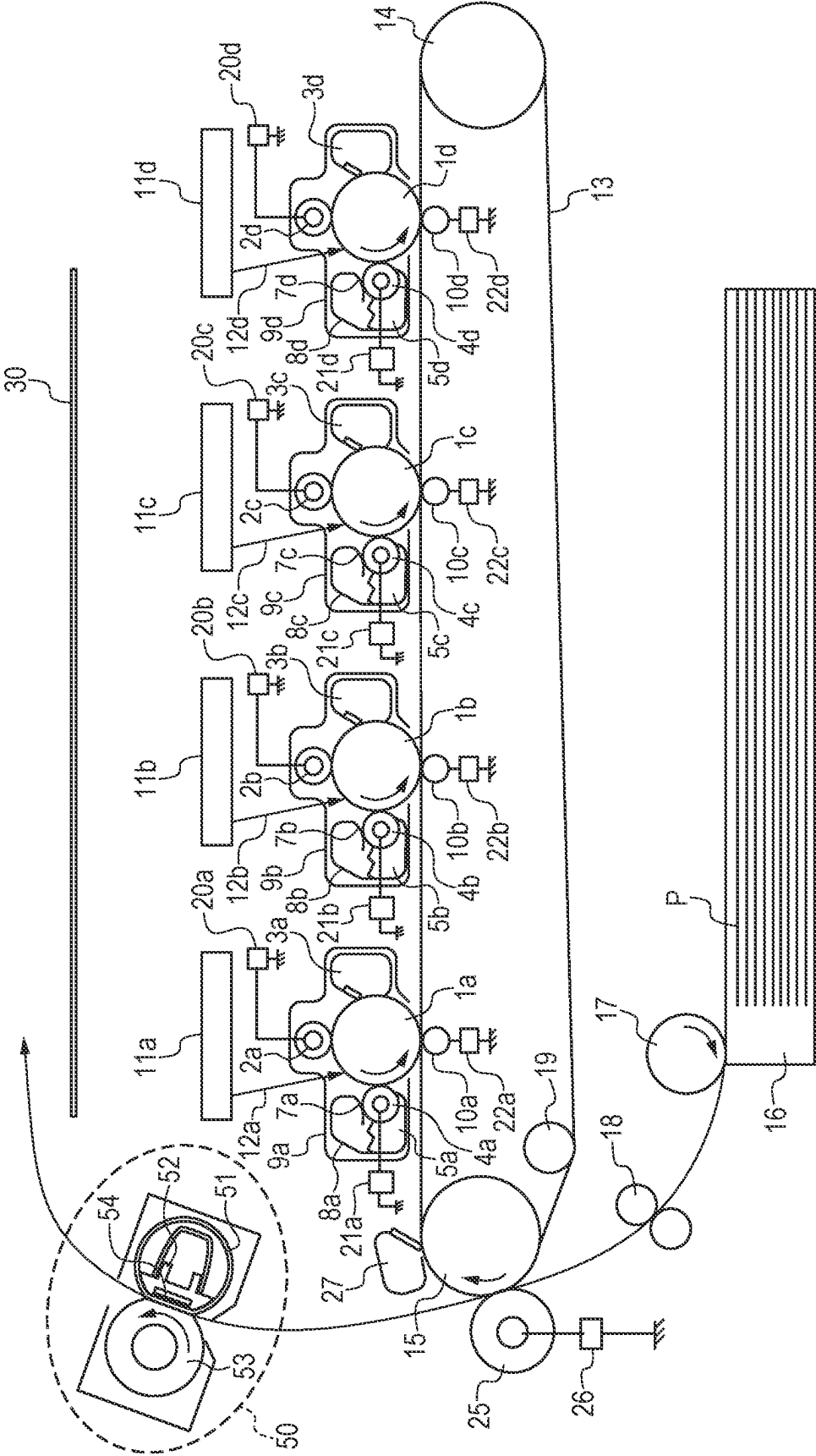


FIG. 2

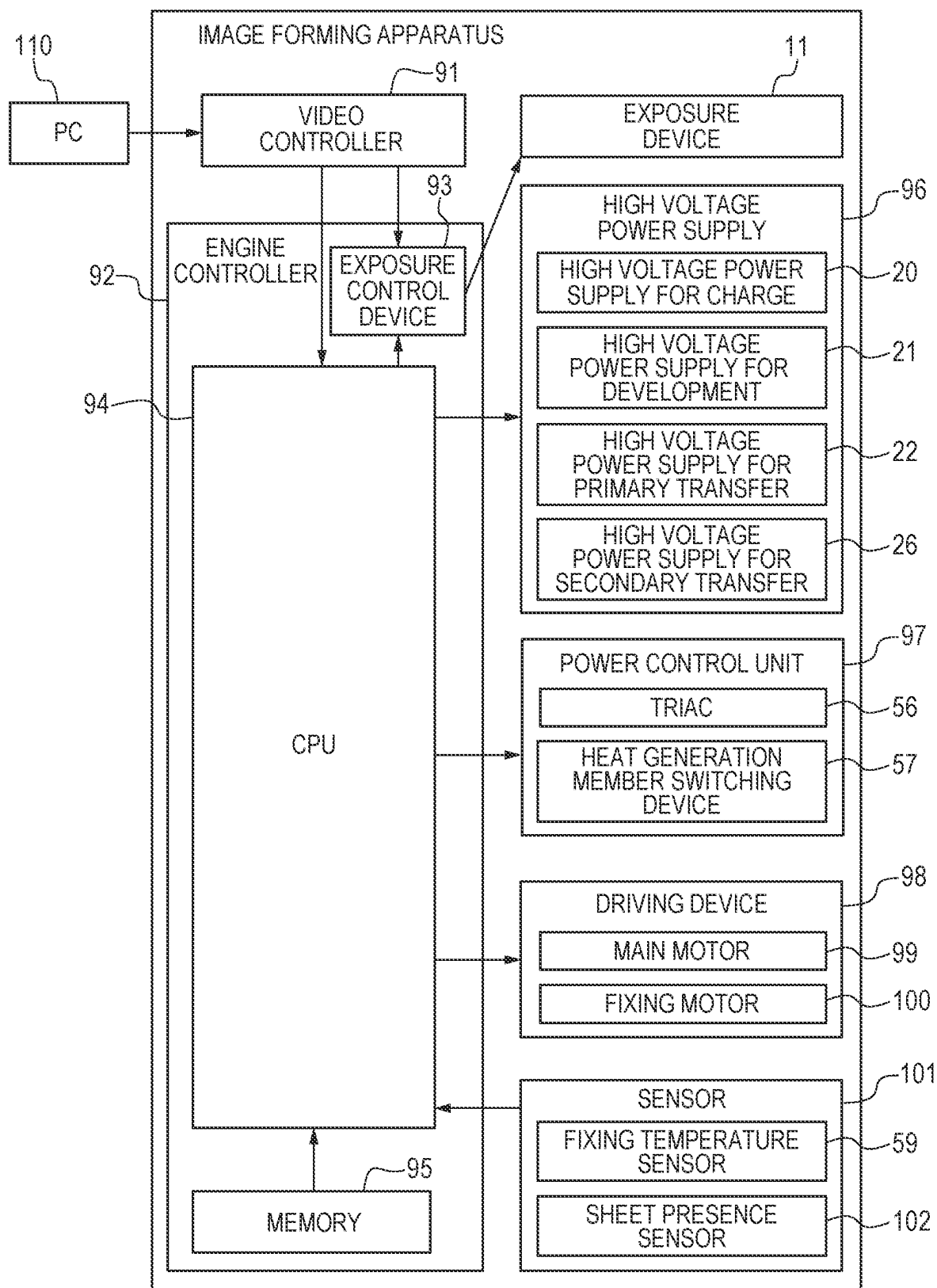


FIG. 3A

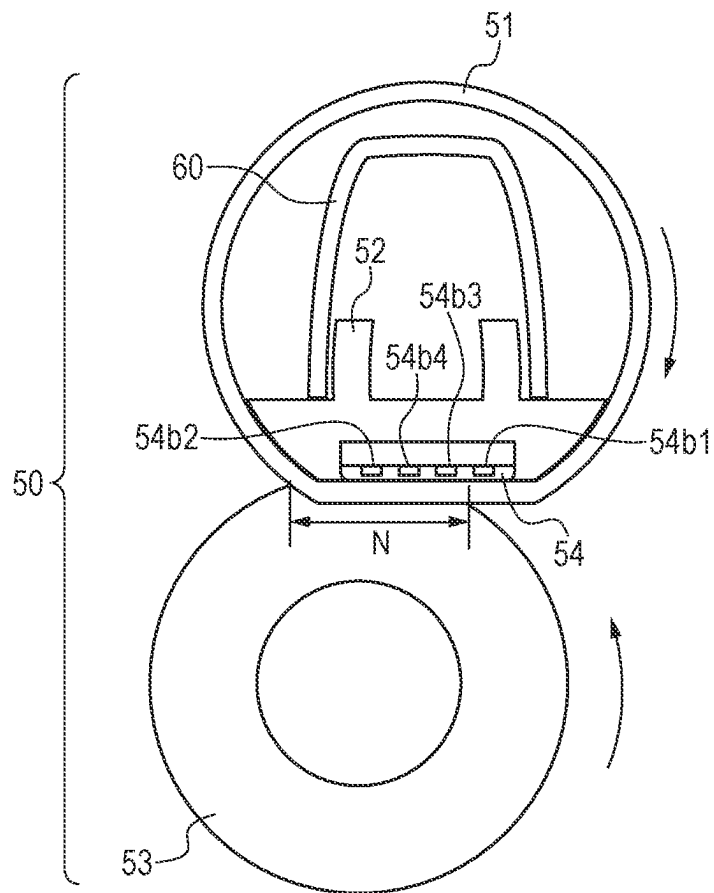


FIG. 3B

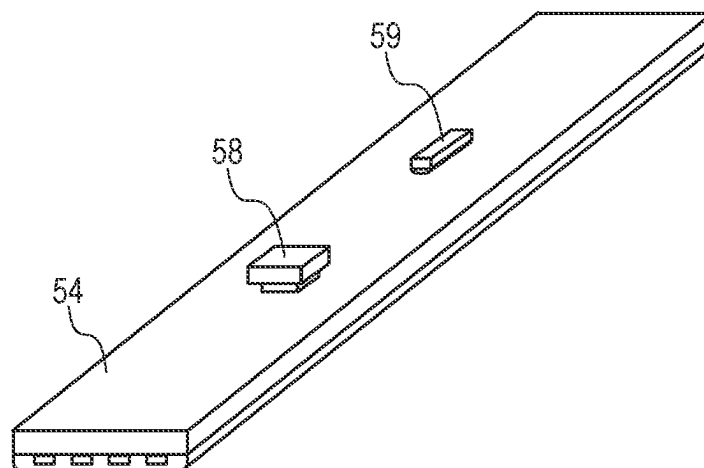


FIG. 4

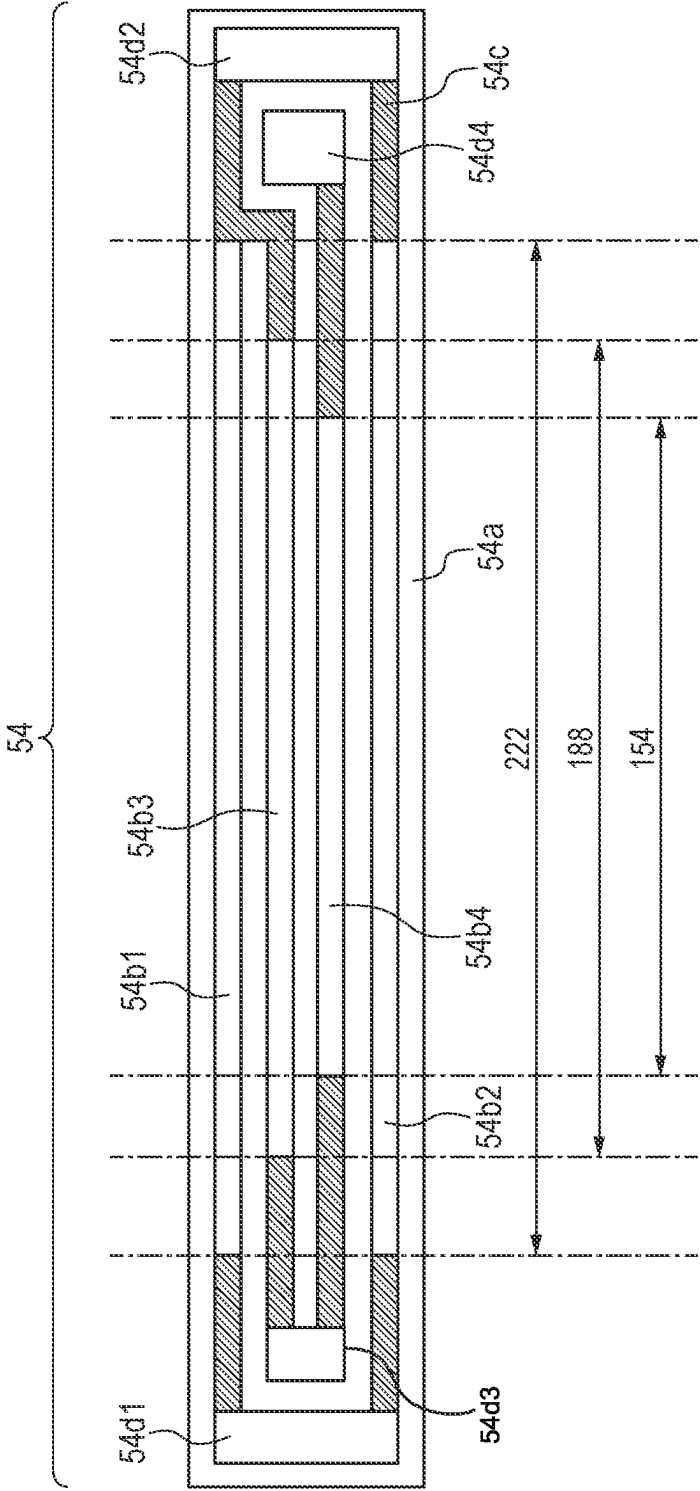


FIG. 5

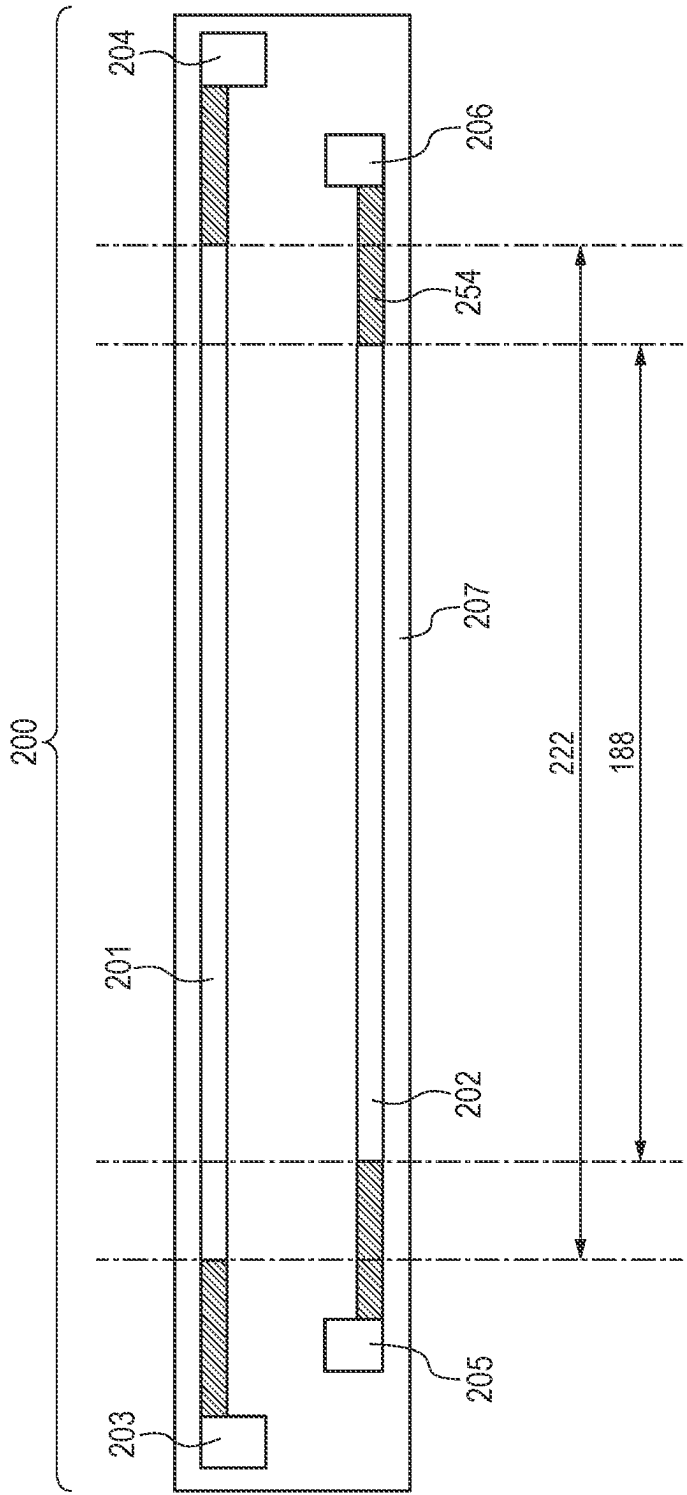


FIG. 6A

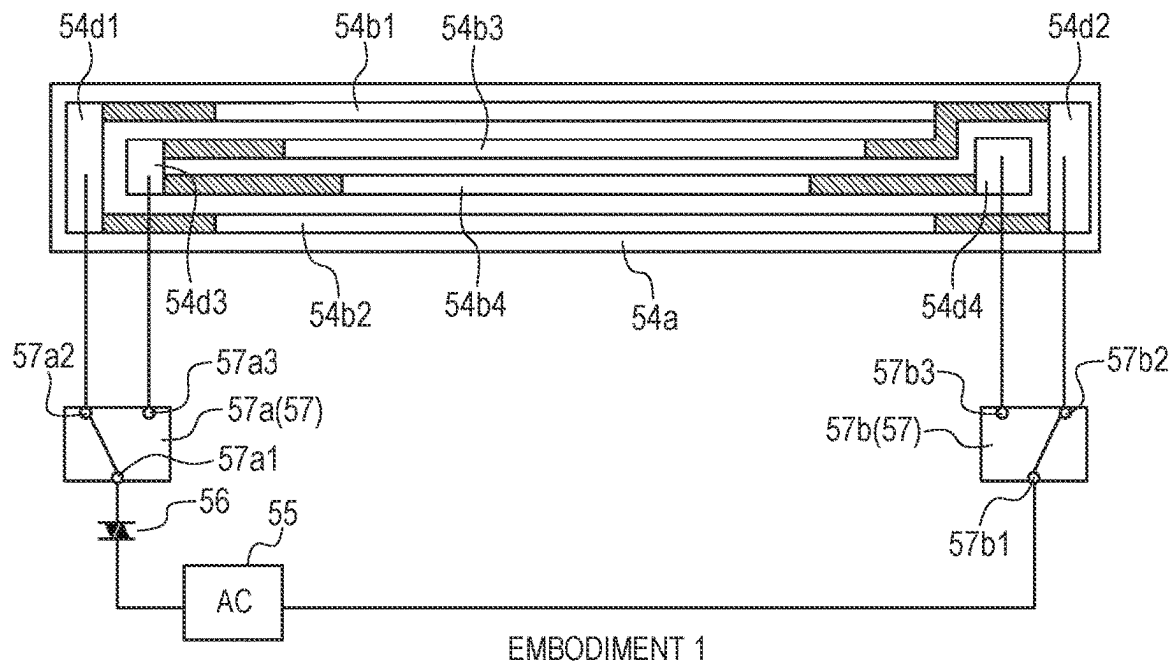


FIG. 6B

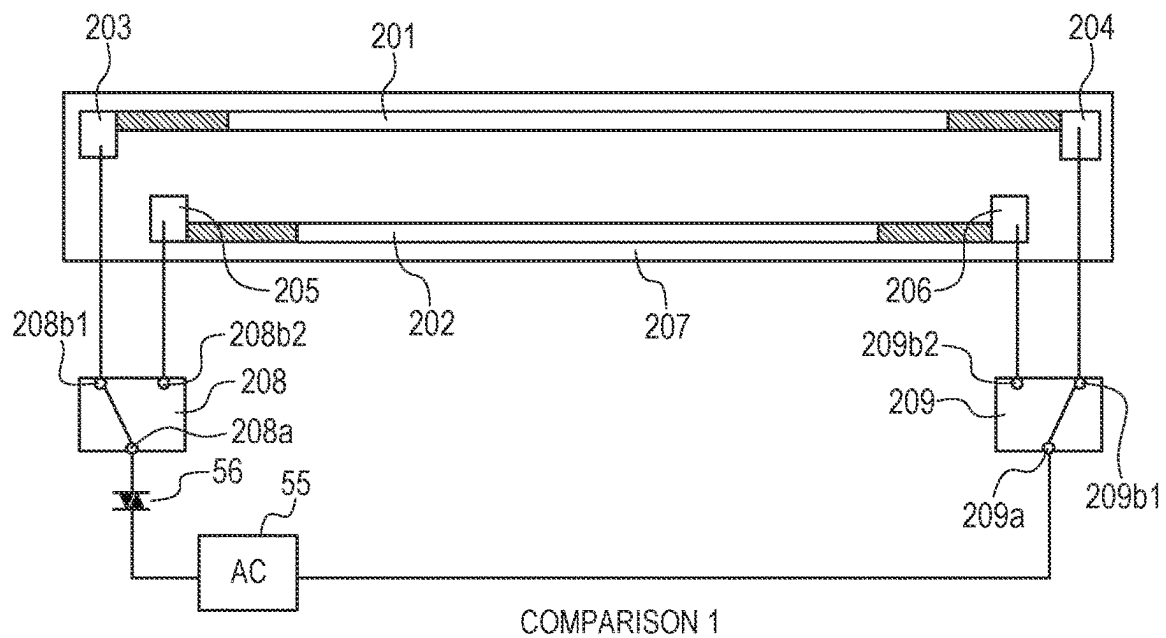


FIG. 7

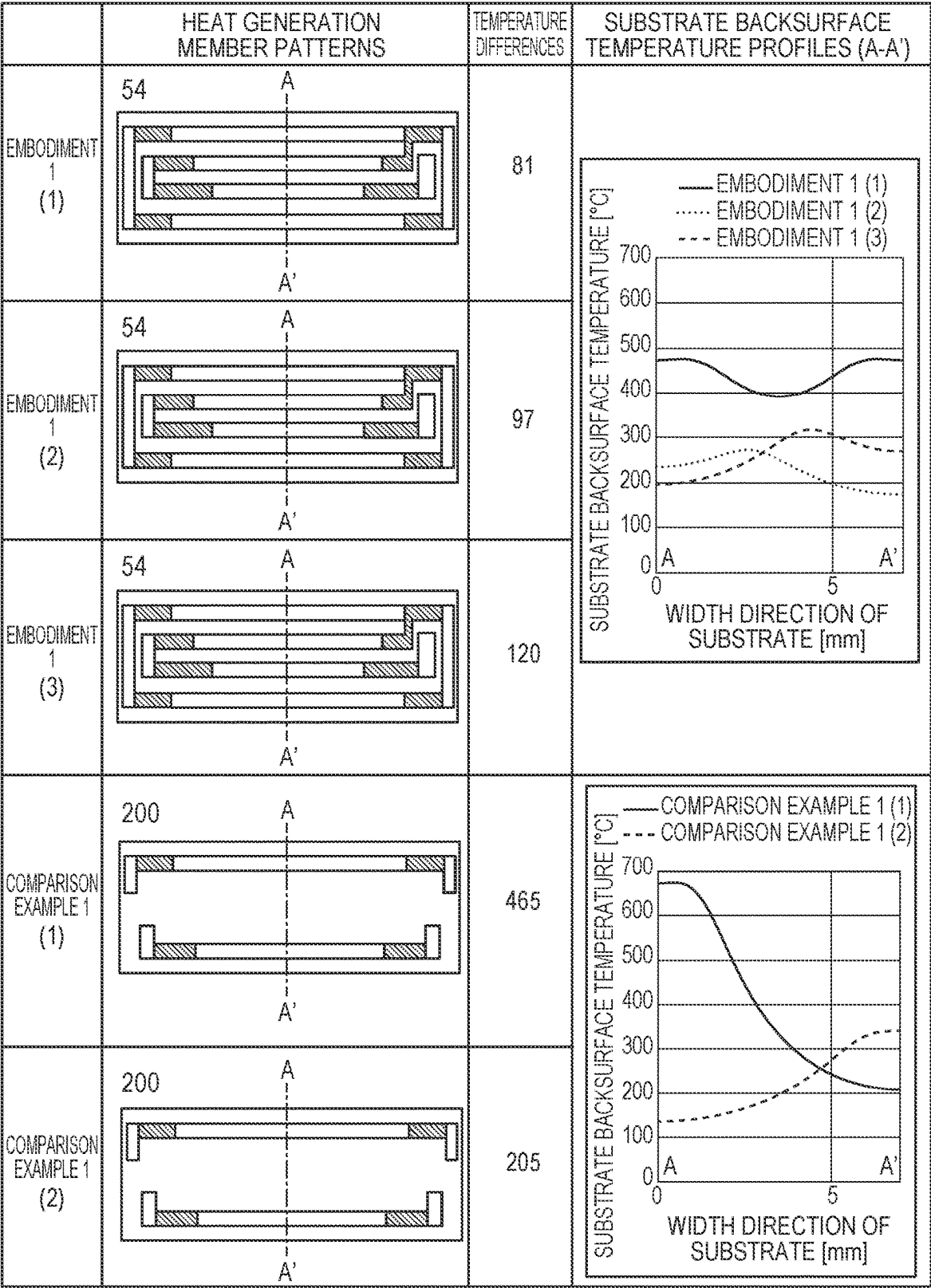


FIG. 8

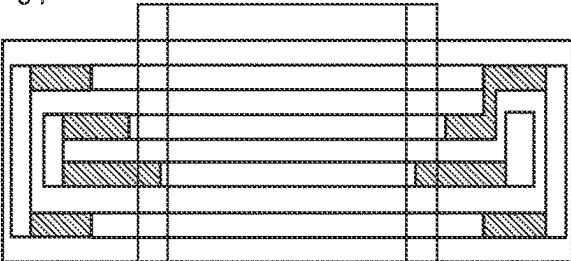
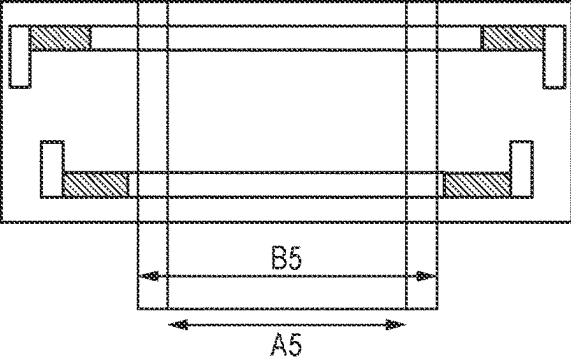
	HEAT GENERATION MEMBER PATTERNS	B5 MAXIMUM PRODUCTIVITY	A5 MAXIMUM PRODUCTIVITY
EMBODIMENT 1	54 	39	46
COMPARISON EXAMPLE 1	200 	39	16

FIG. 9A

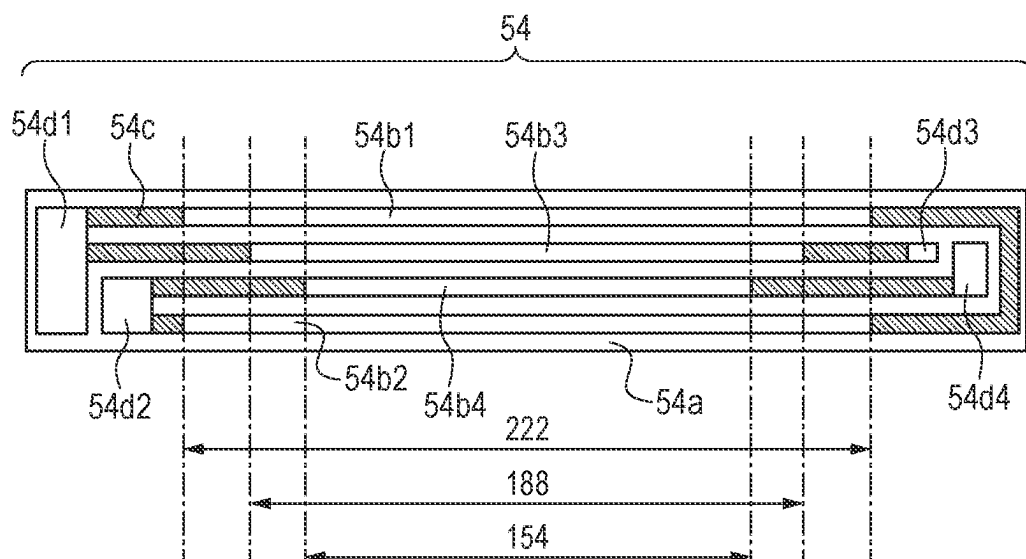


FIG. 9B

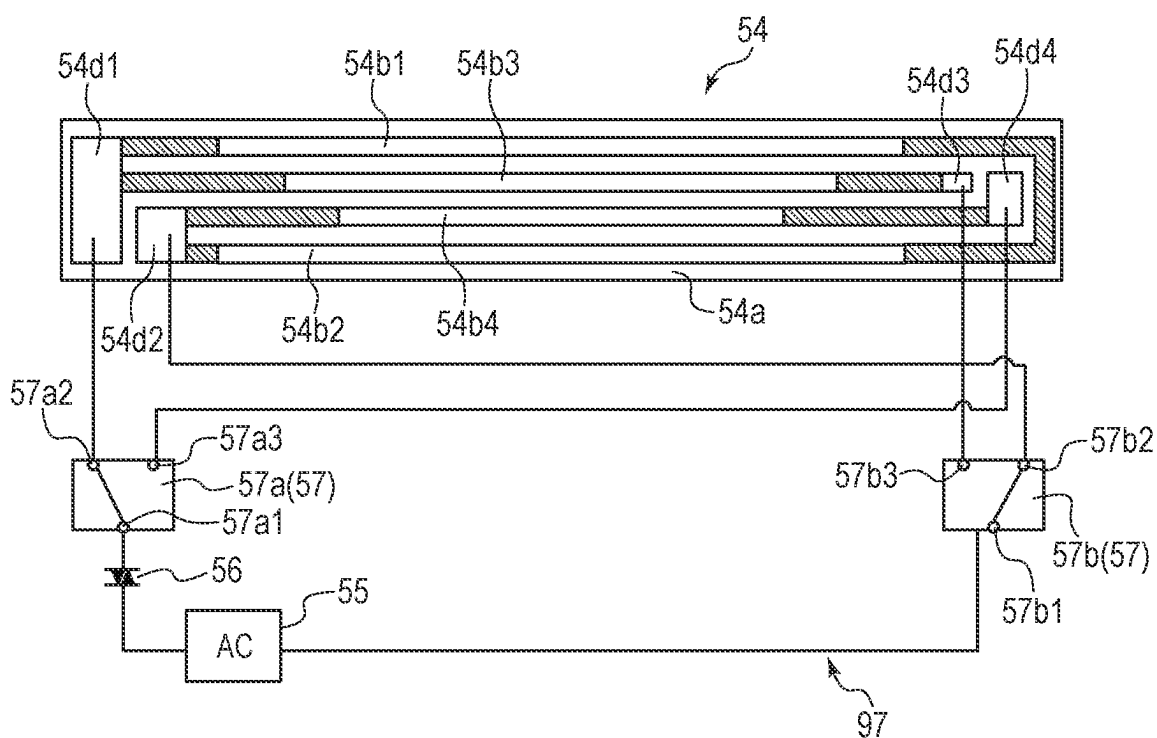


FIG. 10

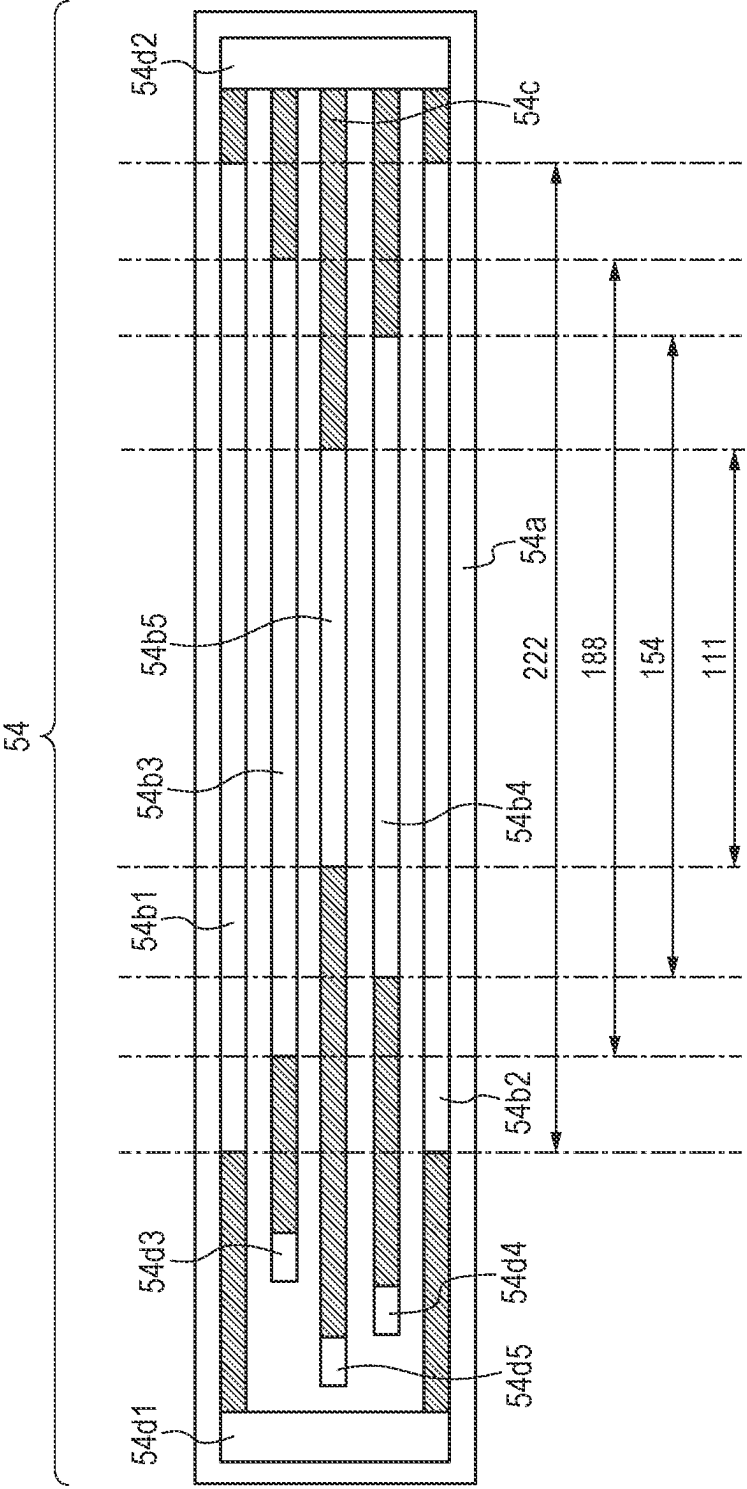


FIG. 11

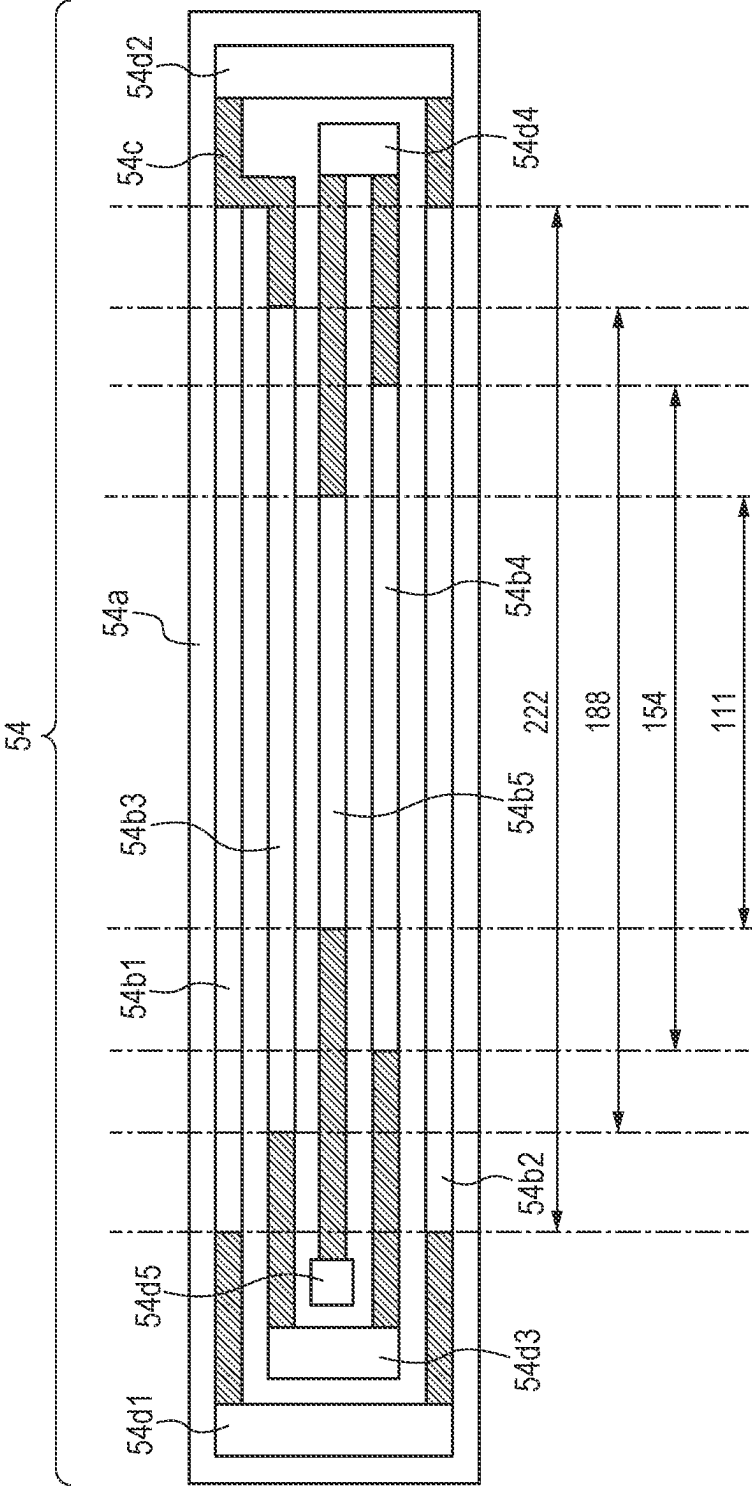


FIG. 12

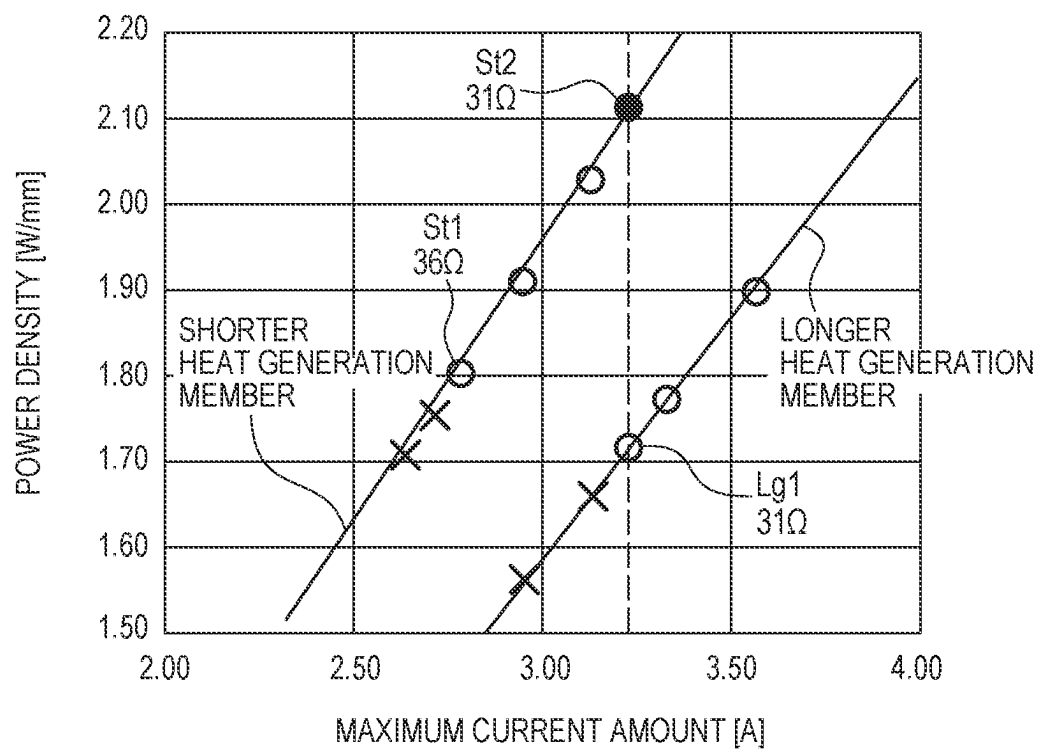


FIG. 13A

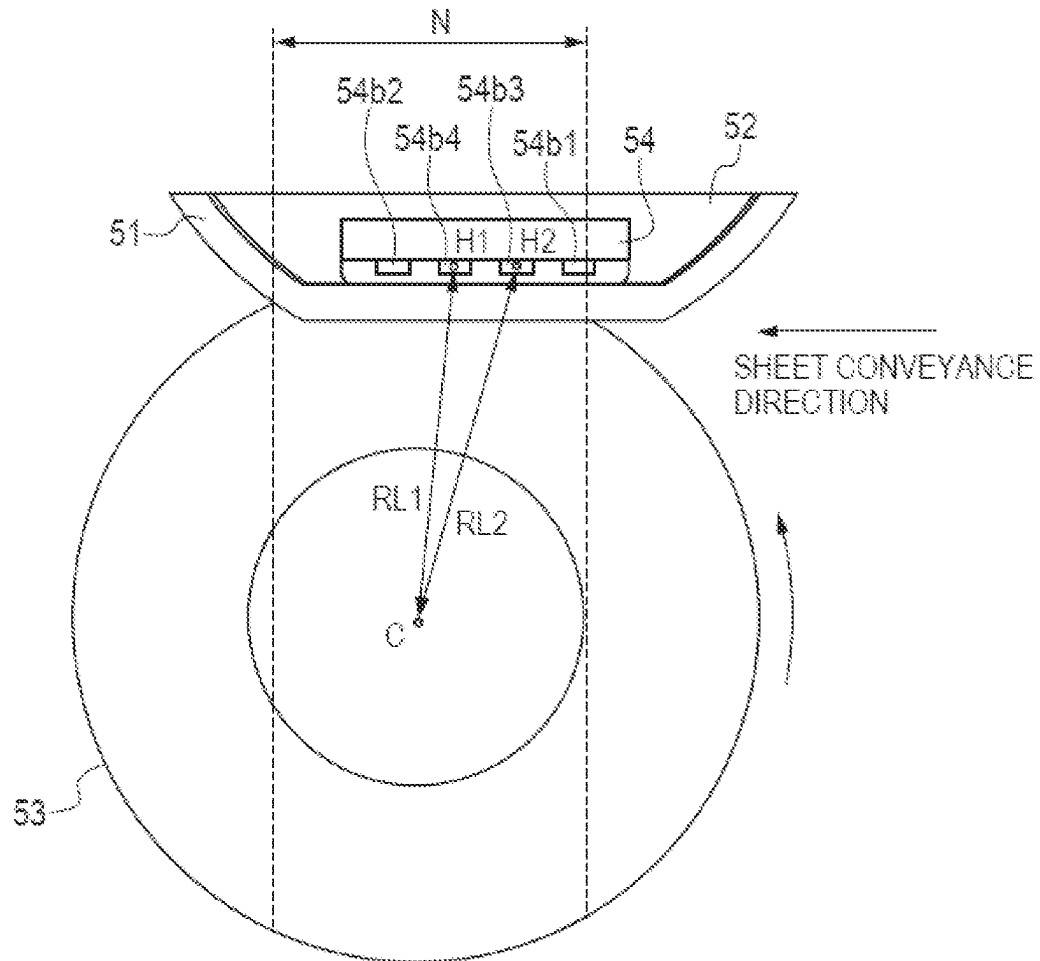
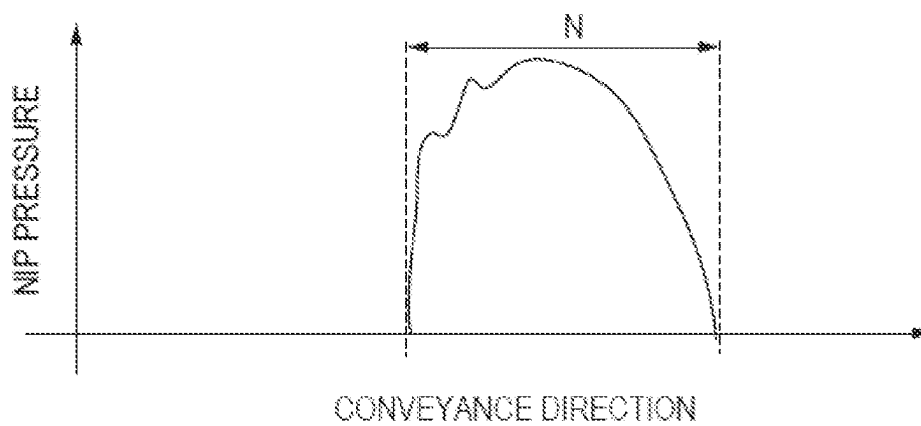


FIG. 13B



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HEATER INCLUDING A PLURALITY OF HEAT GENERATION MEMBERS, FIXING APPARATUS, AND IMAGE FORMING APPARATUS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a Continuation of U.S. patent application Ser. No. 17/883,799, filed on Aug. 9, 2022, which is a Continuation of U.S. patent application Ser. No. 17/352,770, filed on Jun. 21, 2021, issued as U.S. Pat. No. 11,442,385 on Sep. 13, 2022, which is a Continuation of U.S. patent application Ser. No. 16/744,669, filed on Jan. 16, 2020, issued as U.S. Pat. No. 11,073,778 on Jul. 27, 2021, which claims priority to Japanese Patent Application No. 2019-006469, filed on Jan. 18, 2019, the entire disclosures of which are hereby incorporated by reference herein.

BACKGROUND OF THE INVENTION

Field of the Invention

The present invention relates to a heater, a fixing apparatus, and an image forming apparatus, and particularly relates to a fixing apparatus and a heater in an image forming apparatus utilizing an electrophotography recording system, such as a laser printer, a copying machine and a facsimile.

Description of the Related Art

A fixing apparatus heats and fixes, to a paper, an unfixed toner image on the paper by using a heating member that includes a heat generation member having the almost same width (hereinafter referred to as the maximum width) as the maximum paper width that is able to be conveyed (hereinafter referred to as sheet feeding) in a nip portion. On the other hand, the paper sizes used by a user are varied in size, such as A4, B5 and A5. In a case where an A4 size sheet having a wide width is used, since the paper passes through an entire area (hereinafter referred to as a heating area) heated by the heating member including the heat generation member with the maximum width, the heating member and the fixing apparatus maintain a uniform temperature in the entire area. On the other hand, in a case where an A5 paper with a narrow width is used, the paper does not necessarily pass through the entire heating area of the heating member including the heat generation member having the maximum width. That is, although the A5 paper passes through a part of the heating area, the A5 paper does not pass through a part of the heating area. In an area (hereinafter referred to as the sheet feeding area) through which a paper passed in the heating area, since heat is taken by the paper, the temperature is low. On the other hand, in an area (hereinafter referred to as a non-sheet feeding area) through which a paper did not pass in the heating area, since heat is not taken by the paper, the temperature becomes high (temperature rise). There is a possibility of generating adverse image effects due to the temperature rise in this non-sheet feeding area. Therefore, for a paper with a narrow width, the temperature rise in the non-sheet feeding area is suppressed in advance by control that reduces the productivity. In order to suppress this reduction of productivity, for example, in Japanese Patent Application Laid-Open No. 2000-162909, a heat generation member having a wide width and a heat generation member having a narrow width are provided in a heating member, and the heat generation member with the

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narrow width is used when feeding a paper with a narrow width. Accordingly, the temperature rise of the non-sheet feeding area can be reduced, and high productivity can be maintained.

However, in a case where an unexpected circumstance is assumed in which a part of an apparatus breaks down, and power is excessively supplied to one of the heat generation members, there is a possibility that a substrate of the heating member (hereinafter referred to as the heating member substrate) is greatly deformed due to a rapid temperature rise of the heating member. When the temperature of the heating member substrate is partially and greatly increased, a portion having a great temperature rise and a portion having a small temperature rise are generated. In the portion having the great temperature rise, the heating member substrate is greatly extended. On the other hand, in the portion having the small temperature rise, the heating member substrate is hardly extended. Depending on the difference in the extension that differs for each portion of the heating member substrate, a distortion (heat stress) will occur in the heating member substrate. The greater the temperature rise or the temperature gradient generated in the heating member substrate, the greater the distortion (heat stress) generated in the heating member substrate will become.

SUMMARY OF THE INVENTION

One aspect of the present invention is a heater including a substrate, a first heat generation member, a second heat generation member having a length substantially the same in a longitudinal direction as a length of the first heat generation member, a third heat generation member having a length shorter than lengths of the first heat generation member and the second heat generation member in the longitudinal direction, and a fourth heat generation member having a length shorter than length of the third heat generation member in the longitudinal direction, wherein the first heat generation member, the second heat generation member, the third heat generation member and the fourth heat generation member are arranged on the substrate, the first heat generation member is arranged at one end of the substrate in a width direction, the second heat generation member is arranged at another end of the substrate in the width direction, to be symmetrical with the first heat generation member, and the third heat generation member and the fourth heat generation member are arranged between the first heat generation member and the second heat generation member in the width direction of the substrate.

Another aspect of the present invention is a heater including a first heat generation member, a second heat generation member, a third heat generation member having a length shorter than the first heat generation member and the second heat generation member in a longitudinal direction, a fourth heat generation member having a length shorter than the third heat generation member in the longitudinal direction, a first contact to which one ends of the first heat generation member and the second heat generation member are electrically connected, a second contact to which another ends of the first heat generation member and the second heat generation member are electrically connected, a third contact to which another end of the third heat generation member and one end of the fourth heat generation member are electrically connected; and a fourth contact to which another end of the fourth heat generation member is electrically connected.

A further aspect of the present invention is a fixing apparatus for fixing an unfixed toner image carried by a

recording material, the fixing apparatus including a heater including a substrate, a first heat generation member, a second heat generation member having a length substantially a same in a longitudinal direction as a length of the first heat generation member, a third heat generation member having a length shorter than lengths of the first heat generation member and the second heat generation member in the longitudinal direction, and a fourth heat generation member having a length shorter than length of the third heat generation member in the longitudinal direction, wherein the first heat generation member, the second heat generation member, the third heat generation member and the fourth heat generation member are arranged on the substrate, the first heat generation member is arranged at one end of the substrate in a width direction, the second heat generation member is arranged at another end of the substrate in the width direction, to be symmetrical with the first heat generation member, and the third heat generation member and the fourth heat generation member are arranged between the first heat generation member and the second heat generation member in the width direction of the substrate, a first rotary member heated by the heater, and a second rotary member forming a nip portion with the first rotary member.

A still further aspect of the present invention is a fixing apparatus for fixing an unfixed toner image carried by a recording material, the fixing apparatus including a heater having a first heat generation member, a second heat generation member, a third heat generation member having a length shorter than the first heat generation member and the second heat generation member in a longitudinal direction, a fourth heat generation member having a length shorter than the third heat generation member in the longitudinal direction, a first contact to which one ends of the first heat generation member and the second heat generation member are electrically connected, a second contact to which another ends of the first heat generation member and the second heat generation member are electrically connected, a third contact to which another end of the third heat generation member and one end of the fourth heat generation member are electrically connected, and a fourth contact to which another end of the fourth heat generation member is electrically connected.

A still further aspect of the present invention is an image forming apparatus including an image forming unit configured to form an unfixed toner image on a recording material, and a fixing apparatus for fixing an unfixed toner image carried by a recording material, the fixing apparatus including a heater including a substrate, a first heat generation member, a second heat generation member having a length substantially a same in a longitudinal direction as a length of the first heat generation member, a third heat generation member having a length shorter than lengths of the first heat generation member and the second heat generation member in the longitudinal direction, and a fourth heat generation member having a length shorter than length of the third heat generation member in the longitudinal direction, wherein the first heat generation member, the second heat generation member, the third heat generation member and the fourth heat generation member are arranged on the substrate, the first heat generation member is arranged at one end of the substrate in a width direction, the second heat generation member is arranged at another end of the substrate in the width direction, to be symmetrical with the first heat generation member, and the third heat generation member and the fourth heat generation member are arranged between the first heat generation member and the second heat generation member in the width direction of the substrate, a first rotary

member heated by the heater, and a second rotary member forming a nip portion with the first rotary member, wherein the fixing apparatus fixes the unfixed toner image to the recording material.

A still further aspect of the present invention is an image forming apparatus including an image forming unit configured to form an unfixed toner image on a recording material, and a fixing apparatus for fixing an unfixed toner image carried by a recording material, the fixing apparatus including a heater having a first heat generation member, a second heat generation member, a third heat generation member having a length shorter than the first heat generation member and the second heat generation member in a longitudinal direction, a fourth heat generation member having a length shorter than the third heat generation member in the longitudinal direction, a first contact to which one ends of the first heat generation member and the second heat generation member are electrically connected, a second contact to which another ends of the first heat generation member and the second heat generation member are electrically connected, a third contact to which another end of the third heat generation member and one end of the fourth heat generation member are electrically connected, and a fourth contact to which another end of the fourth heat generation member is electrically connected, wherein the fixing apparatus fixes the unfixed toner image to the recording material.

Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a general configuration diagram of an image forming apparatus of Embodiments 1 to 3.

FIG. 2 is a control block diagram of the image forming apparatus of Embodiments 1 to 3.

FIG. 3A and FIG. 3B are diagrams illustrating a fixing apparatus and a heater of Embodiments 1 to 3.

FIG. 4 is a diagram illustrating the heater of Embodiment 1.

FIG. 5 is a diagram illustrating the heater of Comparison Example 1 for comparison with Embodiment 1.

FIG. 6A is a diagram illustrating electric power supply to the heater of Embodiment 1. FIG. 6B is a diagram illustrating the electric power supply to the heater of Comparison Example 1.

FIG. 7 is a diagram illustrating a comparison verification result 1 of Embodiment 1 and Comparison Example 1.

FIG. 8 is a diagram illustrating a comparison verification result 2 of Embodiment 1 and Comparison Example 1.

FIG. 9A and FIG. 9B are diagrams illustrating modifications of the heater of Embodiment 1.

FIG. 10 is a diagram illustrating a modification of the heater of Embodiment 1.

FIG. 11 is a diagram illustrating a modification of the heater of Embodiment 1.

FIG. 12 is a graph illustrating the relationship between the maximum current amount and the power density of Embodiment 2.

FIG. 13A illustrates a cross-sectional view of a fixing apparatus of Embodiment 3. FIG. 13B is a graph illustrating the nip pressure corresponding to the cross-sectional view of the fixing apparatus of Embodiment 3.

DESCRIPTION OF THE EMBODIMENTS

Referring to the drawings, embodiments of the present invention will be described below. In the following embodi-

ments, letting a paper pass through a fixation nip portion will be referred to as sheet feeding. Additionally, in the area in which the heat generation member is generating heat, the area through which a paper is not fed is referred to as the non-sheet feeding area (or the non-sheet feeding portion), and the area through which a paper is fed is referred to as the sheet feeding area (or the sheet feeding portion). Further, the phenomenon in which the temperature in the non-sheet feeding area becomes higher compared with that in the sheet feeding area is referred to as the non-sheet feeding portion temperature rise.

Embodiment 1

[Image Forming Apparatus]

FIG. 1 is a configuration diagram illustrating a color image forming apparatus of the in-line system, which is an example of an image forming apparatus carrying a fixing apparatus of Embodiment 1. The operation of the color image forming apparatus of the electrophotography system will be described by using FIG. 1. Note that it is assumed that a first station is a station for toner image formation of a yellow (Y) color, and a second station is a station for toner image formation of a magenta (M) color. Additionally, it is assumed that a third station is a station for toner image formation of a cyan (C) color, and a fourth station is a station for toner image formation of a black (K) color.

In the first station, a photosensitive drum 1a, which is an image carrier, is an OPC photosensitive drum. The photosensitive drum 1a is formed by stacking, on a metal cylinder, a plurality of layers of functional organic materials including a carrier generation layer exposed and generates an electric charge, a charge transport layer transporting the generated electric charge, etc., and the outermost layer has a low electric conductivity and is almost insulated. A charge roller 2a, which is a charging unit, abuts the photosensitive drum 1a, and uniformly charges a surface of the photosensitive drum 1a while performing following rotation with the rotation of the photosensitive drum 1a. The voltage superimposed with one of a DC voltage and an AC voltage is applied to the charge roller 2a, and when an electric discharge occurs in minute air gaps on the upstream side and the downstream side of a rotation direction from a nip portion between the charge roller 2a and the surface of the photosensitive drum 1a, the photosensitive drum 1a is charged. A cleaning unit 3a is a unit that cleans a toner remaining on the photosensitive drum 1a after the transfer, which will be described later. A development unit 8a, which is a developing unit, includes a developing roller 4a, a nonmagnetic monocomponent toner 5a and a developer application blade 7a. The photosensitive drum 1a, the charge roller 2a, the cleaning unit 3a and the development unit 8a form an integral-type process cartridge 9a that can be freely attached to and detached from the image forming apparatus.

An exposure device 11a, which is an exposing unit, includes one of a scanner unit scanning a laser beam with a polygon mirror, and an LED (light emitting diode) array, and irradiates a scanning beam 12a modulated based on an image signal on the photosensitive drum 1a. Additionally, the charge roller 2a is connected to a high voltage power supply for charge 20a, which is a voltage supplying unit to the charge roller 2a. The developing roller 4a is connected to a high voltage power supply for development 21a, which is a voltage supplying unit to the developing roller 4a. A primary transfer roller 10a is connected to a high voltage power supply for primary transfer 22a, which is a voltage supplying unit to the primary transfer roller 10a. The first

station is configured as described above, and the second, third and fourth stations are also configured in the same manner. For the other stations, the identical numerals are assigned to the components having the identical functions as those of the first station, and b, c and d are assigned as the subscripts of the numerals for the respective stations. Note that, in the following description, the subscripts a, b, c and d are omitted, except for a case where a specific station is described.

An intermediate transfer belt 13 is supported by three rollers, i.e., a secondary transfer opposing roller 15, a tension roller 14, and an auxiliary roller 19, as its stretching members. The force in the direction of stretching the intermediate transfer belt 13 is applied only to the tension roller 14 by a spring, and a suitable tension force for the intermediate transfer belt 13 is maintained. The secondary transfer opposing roller 15 is rotated in response to the rotation drive from a main motor (not illustrated), and the intermediate transfer belt 13 wound around the outer circumference is rotated. The intermediate transfer belt 13 moves at substantially the same speed in a forward direction (for example, the clockwise direction in FIG. 1) with respect to the photosensitive drums 1a to 1d (for example, rotated in the counter clockwise direction in FIG. 1). Additionally, the intermediate transfer belt 13 is rotated in an arrow direction (the clockwise direction), and the primary transfer roller 10 is arranged on the opposite side of the photosensitive drum 1 across the intermediate transfer belt 13, and performs the following rotation with the movement of the intermediate transfer belt 13. The position at which the photosensitive drum 1 and the primary transfer roller 10 abut each other across the intermediate transfer belt 13 is called a primary transfer position. The auxiliary roller 19, the tension roller 14 and the secondary transfer opposing roller 15 are electrically grounded. Note that, also in the second to fourth stations, since primary transfer rollers to 10d are configured in the same manner as the primary transfer roller 10a of the first station, a description will be omitted.

Next, the image forming operation of the image forming apparatus of Embodiment 1 will be described. An image forming apparatus starts the image forming operation, when a print command is received in a standby state. The photosensitive drum 1, the intermediate transfer belt 13, etc. start rotation in the arrow direction at a predetermined process speed by the main motor (not illustrated). The photosensitive drum 1a is uniformly charged by the charge roller 2a to which the voltage is applied by the high voltage power supply for charge 20a, and subsequently, an electrostatic latent image according to image information is formed by the scanning beam 12a irradiated from the exposure device 11a. A toner 5a in the development unit 8a is charged in negative polarity by the developer application blade 7a, and is applied to the developing roller 4a. Then, a predetermined developing voltage is supplied to the developing roller 4a by the high voltage power supply for development 21a. When the photosensitive drum 1a is rotated, and the electrostatic latent image formed on the photosensitive drum 1a reaches the developing roller 4a, the electrostatic latent image is visualized when the toner of negative polarity adheres, and a toner image of the first color (for example, Y (yellow)) is formed on the photosensitive drum 1a. The respective stations (process cartridges 9b to 9d) of the other colors M (magenta), C (cyan) and K (black) are also similarly operated. An electrostatic latent image is formed on each of the photosensitive drums 1a to 1d by exposure, while delaying a writing signal from a controller (not illustrated) with a fixed timing, according to the distance between the primary

transfer positions of the respective colors. ADC high voltage having the reverse polarity to that of the toner is applied to each of the primary transfer rollers **10a** to **10d**. With the above-described processes, toner images are sequentially transferred to the intermediate transfer belt **13** (hereinafter referred to as the primary transfer), and a multi toner image is formed on the intermediate transfer belt **13**.

Thereafter, according to imaging of the toner image, a paper P that is a recording material loaded in a cassette **16** is fed (picked up) by a sheet feeding roller **17** rotated and driven by a sheet feeding solenoid (not illustrated). The fed paper P is conveyed to a registration roller (hereinafter referred to as the resist roller) **18** by a conveyance roller. The paper P is conveyed by the resist roller **18** to a transfer nip portion, which is an abutting portion between the intermediate transfer belt **13** and a secondary transfer roller **25**, in synchronization with the toner image on the intermediate transfer belt **13**. The voltage having the reverse polarity to that of the toner is applied to the secondary transfer roller **25** by a high voltage power supply for secondary transfer **26**, and the four-color multi toner image carried on the intermediate transfer belt **13** is collectively transferred onto the paper P (onto the recording material) (hereinafter referred to as the secondary transfer). The members (for example, the photosensitive drum **1**) that have contributed to the formation of the unfixed toner image on the paper P function as an image forming unit. On the other hand, after completing the secondary transfer, the toner remaining on the intermediate transfer belt **13** is cleaned by a cleaning unit **27**. The paper P to which the secondary transfer is completed is conveyed to a fixing apparatus which is a fixing unit, and is discharged to a discharge tray **30** as an image formed matter (a print, a copy) in response to fixing of the toner image. A film **51** of the fixing apparatus **50**, a nip forming member **52**, a pressure roller **53** and a heater **54** will be described later.

[Block Diagram of Image Forming Apparatus]

FIG. 2 is a block diagram for describing the operation of the image forming apparatus, and referring to this drawing, the print operation of the image forming apparatus will be described. APC **110**, which is a host computer, outputs a print command to a video controller **91** inside the image forming apparatus, and plays the role of transferring image data of a printing image to the video controller **91**.

The video controller **91** converts the image data from the PC **110** into exposure data, and transfers it to an exposure control device **93** inside an engine controller **92**. The exposure control device **93** is controlled from a CPU **94**, and performs turning on and off of exposure data, and control of the exposure device **11**. The CPU **94**, which is a control unit, starts an image forming sequence, when a print command is received.

The CPU **94**, a memory **95**, etc. are mounted in the engine controller **92**, and the operation programmed in advance is performed. The high voltage power supply **96** includes the above-described high voltage power supply for charge **20**, high voltage power supply for development **21**, high voltage power supply for primary transfer **22** and high voltage power supply for secondary transfer **26**. Additionally, a power control unit **97** includes a bidirectional thyristor (hereinafter referred to as the triac) **56**, a heat generation member switching device **57** as a switching unit that exclusively selects a heat generation member supplying power, etc. The power control unit **97** selects the heat generation member that generates heat in the fixing apparatus **50**, and determines the electric energy to be supplied. Additionally, a driving device **98** includes a main motor **99**, a fixing motor **100**, etc. In addition, a sensor **101** includes a fixing tem-

perature sensor **59** that detects the temperature of the fixing apparatus **50**, a sheet presence sensor **102** that has a flag and detects the existence of the paper P, etc., and the detection result of the sensor **101** is transmitted to the CPU **94**. The CPU **94** obtains the detection result of the sensor **101** in the image forming apparatus, and controls the exposure device **11**, the high voltage power supply **96**, the power control unit **97** and the driving device **98**. Accordingly, the CPU **94** performs the formation of an electrostatic latent image, the transfer of a developed toner image, the fixing of a toner image to the paper P, etc., and controls an image formation process in which the exposure data is printed on the paper P as the toner image. Note that the image forming apparatus to which the present invention is applied is not limited to the image forming apparatus having the configuration described in FIG. 1, and may be an image forming apparatus that can print papers P having different widths, and that includes the fixing apparatus **50** including the heater **54**, which will be described later.

[Fixing Apparatus]

FIG. 3A illustrates a cross-section of the fixing apparatus **50** used in Embodiment 1. FIG. 3B illustrates a rear surface of the heater **54**. Referring to FIG. 3A and FIG. 3B, the fixing apparatus **50** will be described below. The fixing apparatus **50** includes a cylindrical film **51**, the pressure roller **53** forming the fixation nip portion N with the film **51**, the heater **54**, which is a heating member, a nip forming member **52** holding the heater **54**, and a stay **60** for maintaining the strength in the longitudinal direction. The film **51**, which is a first rotary member, includes a silicone rubber layer having a film thickness of 200 μm on a polyimide substrate having a film thickness of 50 μm , and a PFA release layer having a film thickness of 20 μm on the silicone rubber layer. The pressure roller **53**, which is a second rotary member, includes an SUM cored bar having an outer diameter of 13 mm, a silicone rubber elastic layer having a film thickness of 3.5 mm on the SUM cored bar, and further includes a PFA release layer having a film thickness of 40 μm on the silicone rubber elastic layer. The pressure roller **53** is rotated by a driving source (not illustrated), and the film **51** performs the following rotation following the driving of the pressure roller **53**.

The heater **54** is provided to contact the inner surface of the film **51**, and is held by the nip forming member **52**, and the inner periphery surface of the film **51** and the top surface of the heater **54** contact each other. Here, in the heater **54**, the surface on which heat generation members **54b1** to **54b4** described later are provided is the top surface, and the surface on which a thermo switch **58**, etc. described later is provided is the rear surface. The stay **60** is pressurized on both ends by a unit that is not illustrated, and the pressurizing force is received by the pressure roller **53** via the nip forming member **52** and the film **51**. Accordingly, a fixation nip portion N at which the film **51** and the pressure roller **53** are pressed and contact each other is formed. The nip forming member **52** is required to have rigidity, heat resistance and thermal insulation properties, and is formed by a liquid crystal polymer. As illustrated in FIG. 3B, the thermo switch **58**, which is a safety element, and the fixing temperature sensor **59** such as a thermistor, which is a temperature detecting unit, contact and are arranged on the rear surface of the heater **54**.

The thermo switch **58** arranged on the rear surface of the heater **54** is, for example, a bimetal thermo switch, and the heater **54** and the thermo switch **58** are electrically connected to each other. When the thermo switch **58** detects that the temperature of the rear surface of the heater **54** has

excessively risen (hereinafter referred to as the excessive temperature rise), a bimetal inside the thermo switch 58 is operated, and the power supplied to the heater 54 can be cut off. The fixing temperature sensor 59 arranged on the rear surface of the heater 54 is a chip resistor-type thermistor. The fixing temperature sensor 59 detects chip resistance, and the detection result is used for the temperature control of the heater 54. The fixing temperature sensor 59 can also detect the excessive temperature rise.

[Heater]

The configuration of the heater 54 of Embodiment 1 is illustrated in FIG. 4, and the details will be described below. A substrate 54a is a plate-like ceramic substrate formed with alumina, etc., and the sizes are, for example, the thickness $t=1$ mm, the width $W=6.3$ mm, and the length $l=280$ mm. The heat generation members 54b1, 54b2, 54b3 and 54b4, a conductor 54c, which is an electric conduction route, and contacts 54d1, 54d2, 54d3 and 54d4 for supplying power are formed on the substrate 54a by a printing process. Hereinafter, the heat generation members 54b1 to 54b4 may be collectively referred to as the heat generation member 54b. In FIG. 4, the heat generation member 54b is indicated by white, the conductor 54c is indicated by hatched lines, and the contacts 54d1 to 54d4 are indicated by black.

The heat generation members 54b are arranged at equal intervals in the order of the heat generation member 54b1 having the longest length (hereinafter also referred to as the width) in the longitudinal direction, the heat generation member 54b3 having the second longest width, the heat generation member 54b4 having the third longest width, and the heat generation member 54b2 having the longest width. The heat generation member 54b1 and the heat generation member 54b2 have substantially the same width. The interval between the heat generation members 54b is, for example, 0.7 mm in Embodiment 1. The sizes of the heat generation members 54b1 and 54b2 are, for example, the thickness $t=10$ μm , the width $W=0.7$ mm, and the length $l=222$ mm in Embodiment 1. The sizes of the heat generation member 54b3 are, for example, the thickness $t=10$ μm , the width $W=0.7$ mm, and the length $l=188$ mm in Embodiment 1. The sizes of the heat generation member 54b4 are, for example, the thickness $t=10$ μm , the width $W=0.7$ mm, and the length $l=154$ mm in Embodiment 1.

The heat generation members 54b1 and 54b2 have the length $l=222$ mm, and are used when printing an A4 size sheet having a width of 210 mm. The heat generation member 54b3 has the length $l=188$ mm, and is used when printing a B5 paper having a width of 182 mm. The heat generation member 54b4 has the length $l=154$ mm, and is used when printing an A5 paper having a width of 148.5 mm.

The heat generation member 54b is a conducting material containing silver and palladium as the main components, and a conducting material containing silver as the main component is used for the conductor 54c and the contacts 54d1 to 54d4. It is assumed that the electrical resistances across both ends of the heat generation members 54b in the longitudinal direction are 20Ω in both the longest heat generation members 54b1 and 54b2, 30Ω in the second longest heat generation member 54b3, and also 30Ω in the third longest heat generation member 54b4. One ends of the longest heat generation members 54b1 and 54b2 are electrically connected by the common contact 54d1, and the other ends are electrically connected by the common contact 54d2. Since the heat generation member 54b1 and the heat generation member 54b2 are connected in parallel, the combined electrical resistance of the longest heat generation

members 54b1 and 54b2 between the contacts 54d1 and 54d2 is 10Ω . In this manner, the combined resistance of the heat generation member 54b1 and the heat generation member 54b2 is 10Ω , and is smaller than the resistance (30Ω) of the heat generation member 54b3 and the heat generation member 54b4.

As described above, the heater 54 includes the heat generation member 54b1, which is a first heat generation member, and the heat generation member 54b2, which is a second heat generation member having substantially the same length as the heat generation member 54b1 in the longitudinal direction. Further, the heater 54 includes the heat generation member 54b3, which is a third heat generation member having a shorter length than the heat generation members 54b1 and 54b2 in the longitudinal direction, and the heat generation member 54b4, which is a fourth heat generation member. The heat generation member 54b1 is provided in one end of the substrate 54a in the width direction, and the heat generation member 54b2 is provided in the other end of the substrate 54a in the width direction. The heat generation members 54b3 and 54b4 are provided between the heat generation member 54b1 and the heat generation member 54b2 in the width direction of the substrate 54a.

Additionally, in Embodiment 1, the contact 54d1, which is a first contact, is the contact to which one ends of the heat generation members 54b1 and 54b2 are electrically connected. The contact 54d2, which is a second contact, is the contact to which the other ends of the heat generation member 54b1, the heat generation member 54b2, and the heat generation member 54b3 are electrically connected. The contact 54d3, which is a third contact, is the contact to which one ends of the heat generation member 54b3 and the heat generation member 54b4 are electrically connected. The contact 54d4, which is a fourth contact, is the contact to which the other end of the heat generation member 54b4 is electrically connected.

Note that, although all the widths W of the heat generation members 54b are the identical width of 0.7 mm in Embodiment 1, there are cases where the selection of material of a conducting material is difficult in order to form the heat generation members 54b having the same width W , depending on the performance required for the fixing apparatus 50. In that case, the widths W of the heat generation members 54b may be different according to the performance required for the fixing apparatus 50.

(Regarding Heat Generation Members 54b1 and 54b2)

The characteristics of the heat generation members 54b1 and 54b2 having the longest width in the above-described heater 54 will be described below. If the fixing apparatus 50 can quickly reach a sufficiently heated fixable state (hereinafter also referred to as the sheet feeding enabled state), a printed matter can be quickly provided to the user. Therefore, the power supply capability of the longest heat generation members 54b1 and 54b2 that can heat the entire area in the longitudinal direction can be maximized, so that any size of paper P may be chosen. The heat generation members 54b3 and 54b4 having the shorter lengths than the longest heat generation members 54b1 and 54b2 in the longitudinal direction are used after the fixing apparatus 50 is sufficiently heated by the longest heat generation members 54b1 and 54b2. Therefore, since the electric energy for fixing a toner image to the paper P at the time of sheet feeding may be supplemented, in a case where the heat generation members 54b3 and 54b4 are used, the heat generation members 54b3 and 54b4 can have lower power supply capability compared

to the high power supply capability of the longest heat generation members **54b1** and **54b2**.

When the longest heat generation members **54b1** and **54b2** have the high power supply capability, it means that the deformation risk of the substrate **54a** is high in a case where power is excessively supplied to the longest heat generation members **54b1** and **54b2** due to an unexpected apparatus failure. In Embodiment 1, the longest heat generation members include the two heat generation members **54b1** and **54b2**, one heat generation member **54b1** is arranged on one end of the substrate **54a** in the width direction, and the other heat generation member **54b2** is arranged on the other end of the substrate **54a** in the width direction. Accordingly, the two longest heat generation members **54b1** and **54b2** are arranged so that they are symmetrical in the width direction of the substrate **54a**.

Further, each of the heat generation members **54b1** and **54b2** is electrically connected to each other by the common contacts **54d1** and **54d2**, and the two heat generation members **54b1** and **54b2** are configured such that power is always supplied substantially at the same time. Accordingly, since the both ends of the heater **54** in the width direction always generate heat when power is supplied to the longest heat generation members **54b1** and **54b2**, the supplied electric energy can be distributed, and the temperature gradient of the substrate **54a** in the width direction can be reduced.

As described above, the fixing apparatus **50** can be made to reach the sheet feeding enabled state in a short time, and even if an unexpected apparatus failure occurs, and results in an excessive power supplying state, the temperature gradient of the substrate **54a** in the width direction can be reduced, and the deformation risk of the substrate **54a** can be reduced.

(Regarding Heat Generation Members **54b3** and **54b4**)

Next, the characteristics of the two kinds of non-longest heat generation members **54b3** and **54b4** will be mentioned below. One ends of the heat generation member **54b3** and the heat generation member **54b4** are electrically connected to the one contact **54d3**. On the other hand, in the heat generation member **54b3** and the heat generation member **54b4**, the other end of the heat generation member **54b3** is electrically connected to the contact **54d2**, and the other end of the heat generation member **54b4** is electrically connected to the contact **54d4**. That is, the heat generation member **54b3** and the heat generation member **54b4** are configured so that either one of them will generate heat.

As described above, the heat generation member **54b3** is used at the time of printing of a B5 paper, and the heat generation member **54b4** is used at the time of printing of an A5 paper. The width (hereinafter referred to as the paper width) of the paper P and the lengths of the heat generation members **54b3** and **54b4** in the longitudinal direction are almost the same length, and the paper P passes through most of the area (hereinafter referred to as the heat generation area) in which the heat generation members **54b3** and **54b4** generate heat. Therefore, since most of the heat generated by the heat generation members **54b3** and **54b4** can be provided to the paper P, the temperature rise in the non-sheet feeding area through which the paper P does not pass can be suppressed. Accordingly, maintaining a high productivity is enabled. Additionally, since the longest heat generation members **54b1** and **54b2** are responsible for heating the fixing apparatus to the sheet feeding enabled state, the non-longest heat generation members **54b3** and **54b4** may supplement the electric energy for fixing a toner image to the paper P at the time of sheet feeding. Therefore, the power supply capability of the non-longest heat generation mem-

bers **54b3** and **54b4** can be reduced, and the degree of temperature rise of the heat generation members **54b3** and **54b4** at the time of malfunction can be reduced.

Additionally, the above-described two kinds of heat generation members **54b3** and **54b4** are arranged between the longest heat generation member **54b1** and the longest heat generation member **54b2**, and the heat generation members **54b3** and **54b4** are arranged close to the center of the substrate **54a** in the width direction as much as possible. Accordingly, the temperature rise can be performed almost equally in either of a first end, which is one end of the substrate **54a** in the width direction, and a second end, which is the other end of the substrate **54a**, and the temperature gradient of the substrate **54a** in the width direction can be reduced.

As described above, the power supply capability of the non-longest heat generation members **54b3** and **54b4** is reduced, and the non-longest heat generation members **54b3** and **54b4** are arranged as symmetrically as possible in the width direction of the substrate **54a**. Accordingly, even an unexpected apparatus failure results in an excessive power supplying state, since the temperature gradient in the width direction of the substrate **54a** can be reduced, the deformation risk of the substrate **54a** can be reduced. Additionally, by making the number of only the longest heat generation members **54b1** and **54b2** that require the high power supply capability two, and the number of the non-longest heat generation members **54b3** and **54b4** one, which is the minimally required number, while considering their symmetry in the width direction, the reduction of the size of the substrate **54a** can be achieved at the same time.

Comparison Examples

FIG. 5 illustrates a heater **200** in Comparison Example 1, and the details of the configuration will be described below. A substrate **207** is a plate-like ceramic substrate formed with alumina, etc., and the sizes are, for example, the thickness $t=1$ mm, the width $W=6.3$ mm, and the length $l=280$ mm. Heat generation members **201** and **202**, a conductor **254**, and contacts **203**, **204**, **205** and **206** are formed on the substrate **207** by a printing process. In FIG. 5, the heat generation members **201** and **202** are indicated by white, the conductor **254** is indicated by hatched lines, and the contacts **203** to **206** are indicated by black.

In the heater **200**, two heat generation members, i.e., the heat generation member **201** having the longest width and the heat generation member **202** having the second longest width, are arranged on the substrate **207** with an interval of 3.5 mm. The sizes of the heat generation member **201** are the thickness $t=10$ μm , the width $W=0.7$ mm, and the length $l=222$ mm. The sizes of the heat generation member **202** are the thickness $t=10$ μm , the width $W=0.7$ mm, and the length $l=188$ mm. The heat generation member **201** is used when printing an A4 (210 mm in the width) paper, and the heat generation member **202** is used when printing a B5 (182 mm) paper. The electrical resistances across both ends of the heat generation members **201** and **202** in the longitudinal direction are 10Ω in the longest heat generation member **201**, and 30Ω in the second longest heat generation member **202**. The both ends of the longest heat generation member **201** are electrically connected to the contacts **203** and **204** via the conductor **254**, and the both ends of the second longest heat generation member **202** are electrically connected to the contacts **205** and **206** via the conductor **254**.

Embodiment 1 and Comparison Example 1

FIG. 6A illustrates a power supplying circuit of Embodiment 1. FIG. 6B illustrates the power supplying circuit of

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Comparison Example 1. The comparison verification in these circuits to which Embodiment 1 and Comparison Example 1 are applied will be described. Each of the power supplying circuit will be described below. In Embodiment 1 of FIG. 6A, the contacts **54d1** to **54d4** are connected to a heat generation member switching device **57** for switching the power supply passages. Note that, since the heat generation member **54b** that generates heat is switched by switching the power supply passages by the heat generation member switching device **57**, the switching of the power supply passages is also expressed as the switching of the heat generation member **54b**. In Embodiment 1, specifically, the heat generation member switching devices **57** are electromagnetic relays **57a** and **57b** having c-contact configurations.

The electromagnetic relay **57a** includes a contact **57a1** connected to a first pole of an AC power supply **55** via a triac **56**, a contact **57a2** connected to the contact **54d1**, and a contact **57a3** connected to the contact **54d3**. The electromagnetic relay **57a** is brought into either one of the states, i.e., the state where the contact **57a1** and the contact **57a2** are connected to each other, and the state where the contact **57a1** and the contact **57a3** are connected to each other, by the control of the engine controller **92**. The electromagnetic relay **57b** includes a contact **57b1** connected to a second pole of the AC power supply **55**, a contact **57b2** connected to the contact **54d2**, and a contact **57b3** connected to the contact **54d4**. The electromagnetic relay **57b** is brought into one of the states, i.e., the state where the contact **57b1** and the contact **57b2** are connected to each other, and the state where the contact **57b1** and the contact **57b3** are connected to each other, by the control of the engine controller **92**.

FIG. 6A illustrates the electromagnetic relays **57a** and **57b** at the time of non-operation, the contact **57a1** and the contact **57a2** are connected to each other in the electromagnetic relay **57a**, and the contact **57b1** and the contact **57b2** are connected to each other in the electromagnetic relay **57b**. Since power is supplied between the contact **54d1** and the contact **54d2** at the time of non-operation of the electromagnetic relays **57a** and **57b**, the longest heat generation members **54b1** and **54b2** generate heat.

In a case where the electromagnetic relays **57a** and **57b** are operated, the contact **57a1** and the contact **57a3** are connected to each other in the electromagnetic relay **57a**, and the contact **57b1** and the contact **57b3** are connected to each other in the electromagnetic relay **57b**. Since power is supplied between the contact **54d3** and the contact **54d4** at the time of operation of the electromagnetic relays **57a** and **57b**, only the heat generation member **54b4** generates heat. In a case where only the electromagnetic relay **57a** is operated, it will be in a state where the contact **57a1** and the contact **57a3** are connected to each other in the electromagnetic relay **57a**, and the contact **57b1** and the contact **57b2** are connected to each other in the electromagnetic relay **57b**. Since power is supplied between the contact **54d3** and the contact **54d2** at the time of operation of only the electromagnetic relay **57a**, only the heat generation member **54b3** generates heat.

In Comparison Example 1 of FIG. 6B, the contacts **203** to **206** are connected to electromagnetic relays **208** and **209** having the c-contact configurations, which are heat generation member switching devices for switching power supply passages. The electromagnetic relay **208** includes a contact **208a** connected to the first pole of the AC power supply **55** via the triac **56**, a contact **208b1** connected to the contact **203**, and a contact **208b2** connected to the contact **205**. The electromagnetic relay **208** is brought into either one of the

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states, i.e., the state where the contact **208a** and the contact **208b1** are connected to each other, and the state where the contact **208a** and the contact **208b2** are connected to each other, by the control of the engine controller **92**. The electromagnetic relay **209** includes a contact **209a** connected to the second pole of the AC power supply **55**, a contact **209b1** connected to the contact **204**, and a contact **209b2** connected to the contact **206**. The electromagnetic relay **209** is brought into either one of the states, i.e., the state where the contact **209a** and the contact **209b1** are connected to each other, and the state where the contact **209a** and the contact **209b2** are connected to each other, by the control of the engine controller **92**.

FIG. 6B illustrates the electromagnetic relays **208** and **209** at the time of non-operation, the contact **208a** and the contact **208b1** are connected to each other in the electromagnetic relay **208**, and the contact **209a** and the contact **209b1** are connected to each other in the electromagnetic relay **209**. Since power is supplied between the contact **203** and the contact **204** at the time of non-operation of the electromagnetic relays **208** and **209**, the longest heat generation member **201** generates heat.

In a case where the electromagnetic relays **208** and **209** are operated, the contact **208a** and the contact **208b2** are connected to each other in the electromagnetic relay **208**, and the contact **209a** and the contact **209b2** are connected to each other in the electromagnetic relay **209**. Since power is supplied between the contact **205** and the contact **206** at the time of operation of the electromagnetic relays **208** and **209**, only the heat generation member **202** generates heat. Note that a contact switch, such as an electromagnetic relay having the a-contact configuration, or an electromagnetic relay having the b-contact configuration may be used for the electromagnetic relay, or a contactless switch, such as a solid state relay (SSR), a photoMOS relay, and a triac, may be used for the electromagnetic relay.

Temperature Gradient of Embodiment 1 and Comparison Example 1

(i) In order to estimate the deformation amount of the substrate at the time when an excessive power is supplied to the heat generation member, the temperature profile of the back surface of the substrate (the position indicated by an A-A' line) after 3 seconds since the power was supplied was measured, in a case where AC voltage of 100V was continued to be supplied to the respective heat generation members of Embodiment 1 and Comparison Example 1. It is shown that the larger the difference between the maximum value and the minimum value of the temperature profile, the higher the deformation risk of the substrate.

FIG. 7 illustrates Embodiment 1, Comparison Example 1, etc. in the first row, and illustrates the heat generation pattern of the heater in the second row. Note that the heat generation members to which power was supplied are indicated by vertical stripes. FIG. 7 illustrates the difference (hereinafter referred to as the temperature difference) between the maximum value and the minimum value of the temperature profile in the third row, and illustrates the temperature profile (substrate back surface temperature profile) of the back surface corresponding to the position indicated by the A-A' line of the substrate in the fourth row. In the graphs of the temperature profile, the horizontal axes represent the width direction (temperature width) [mm] of the substrate, and the vertical axes represent the temperature (substrate back surface temperature) [° C.]. Note that in the diagrams of the heat generation patterns, numerals are omitted for visibility.

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Note that, in the graph of Embodiment 1, Embodiment 1 (1) is represented by a solid line, Embodiment 1 (2) is represented by a dotted line, and Embodiment 1 (3) is represented by a broken line. Additionally, in the graph of Comparison Example 1, Comparison Example 1 (1) is represented by a solid line, and Comparison Example 1 (2) is represented by a broken line.

Additionally, Embodiment 1 (1) represents a case where power is supplied to the two longest heat generation members **54b1** and **54b2** corresponding to an A4 size sheet. Embodiment 1 (2) represents a case where power is supplied to the second longest heat generation member **54b3** corresponding to a B5 paper. Embodiment 1 (3) represents a case where power is supplied to the shortest heat generation member **54b4** corresponding to an A5 paper. Comparison Example 1 (1) represents a case where power is supplied to the longest heat generation member **201** corresponding to an A4 size sheet, and Comparison Example 1 (2) represents a case where power is supplied to the second longest heat generation member **202** corresponding to a B5 paper.

Embodiment 1 (1)

In Embodiment 1 (1), the highest temperature of the back surface of the substrate **54a** reached 472° C. near the heat generation member **54b1** or the heat generation member **54b2**, and the lowest temperature was 391° C. between the two heat generation members **54b1** and **54b2**. The difference between the highest temperature and the lowest temperature was 81° C., and the temperature gradient in the substrate **54a** was small. In the configuration of Embodiment 1 (1), the two longest heat generation members **54b1** and **54b2** are used to distribute the electric energy, and are symmetrically arranged on the both ends of the substrate **54a** in the width direction, and the two heat generation members **54b1** and **54b2** share the common contacts **54d1** and **54d2** to always generate heat at the same time. Accordingly, the temperature gradient generated in the substrate **54a** was able to be reduced.

Embodiment 1 (2)

In Embodiment 1 (2), the highest temperature of the back surface of the substrate **54a** reached 271° C. near the heat generation member **54b3**, and the lowest temperature was 174° C. at one end in the width direction, which is the farther end from the heat generation member **54b3**. The difference between the highest temperature and the lowest temperature was 97° C., and the temperature gradient in the substrate **54a** was small. Since the power supply capability of the second longest heat generation member **54b3** of Embodiment 1 (2) is made to be the minimum value required, and the second longest heat generation member **54b3** is arranged in almost the center of the substrate **54a** in the width direction to be symmetrical with the heat generation member **54b4** as much as possible, the temperature gradient generated in the substrate **54a** was able to be reduced.

Embodiment 1 (3)

In Embodiment 1 (3), the highest temperature of the back surface of the substrate **54a** reached 316° C. near the heat generation member **54b4**, and the lowest temperature was 196° C. at one end in the width direction, which is the farther end from the heat generation member **54b4**. The difference between the highest temperature and the lowest temperature was 120° C. For the same reason as the reason described in

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the Embodiment 1 (2), the temperature gradient generated in the substrate **54a** was able to be reduced.

Comparison Example 1 (1)

In Comparison Example 1 (1), the highest temperature of the back surface of the substrate **207** reached 673° C. near the heat generation member **201**, and the lowest temperature was 208° C. at one end in the width direction, which is the farther end from the heat generation member **201**. The difference between the highest temperature and the lowest temperature was 465° C., and the temperature gradient in the substrate **207** was large. In Comparison Example 1 (1), since the number of the longest heat generation member **201** that gives the maximum power supply capability is one, and the longest heat generation member **201** is arranged at one end of the substrate **207** in the width direction, the increase in the temperature at the one end became large.

Comparison Example 1 (2)

In Comparison Example 1 (2), the highest temperature of the back surface of the substrate **207** reached 341° C. near the heat generation member **202**, and the lowest temperature was 136° C. at one end in the width direction, which is the farther end from the heat generation member **202**. The difference between the highest temperature and the lowest temperature was 205° C., and the temperature gradient in the substrate **207** was large. Since the heat generation member **202** has a low power supply capability compared with the heat generation member **201** of Comparison Example 1 (1), although the temperature gradient is smaller than that in Comparison Example 1 (1), the increase in the temperature at one end became large, since the heat generation member **202** is arranged at the one end of the substrate **207** in the width direction.

From the above, while the maximum temperature difference in Embodiment 1 is 120° C., which is shown in the Embodiment 1 (3), the maximum temperature difference in Comparison Example 1 is 465° C., which is shown in Comparison Example 1 (1), and the temperature difference in Comparison Example 1 is three or more times larger than that in Embodiment 1. The extension of the substrate is large in a portion with a high temperature, and the extension of the substrate is small in a portion with a low temperature, and the substrate is deformed due to the difference in the amount of extension. In Embodiment 1, it was able to confirm that, in any of the heat generation members **54b**, the temperature difference was 120° C. or less, which is sufficiently small compared with that in Comparison Example 1, and the risk of deformation of the substrate **54a** was small. Even if the material of the substrate and the sizes of the substrate are changed, the same effects can be obtained by using the configuration illustrated in the Embodiment 1.

Productivity of Embodiment 1 and Comparison Example 1

(ii) FIG. 8 illustrates the confirmation results of the maximum productivity for a B5 paper and an A5 paper in Embodiment 1 and Comparison Example 1. FIG. 8 illustrates Embodiment 1 and Comparison Example 1 in the first row, and illustrates the patterns of the heat generation member in the second row. The width of a B5 paper and the width of an A5 paper are also illustrated in the heat generation member patterns. FIG. 8 illustrates the maximum productivity at the time when B5 papers are continuously

printed in the third row, and illustrates the maximum productivity at the time when A5 papers are continuously printed in the fourth row.

The conditions for an image forming apparatus and a fixing apparatus at the time of confirming the productivity will be mentioned. A paper P previously printed is herein-
after referred to as the preceding paper, and the subsequent paper printed subsequently to the paper P is hereinafter referred to as the subsequent paper. Additionally, the interval between the bottom end of the preceding paper and the top end of the subsequent paper is hereinafter also referred to as the paper interval. The image process speed of the image forming apparatus is 200 mm/sec, the interval (paper interval) between the preceding paper and the subsequent paper is 50 mm (0.4 second), and papers P having the same size are continuously fed while maintaining the maximum productivity. Sheet feeding is performed by performing the temperature control by the engine controller 92, so that the back surface of the substrate becomes 180° C. by the fixing temperature sensor 59 installed in the back surface of the substrate. As for the papers P, Canon CS680 having the B5 (182 mm in width×257 mm in length×92 μm in thickness, a basis weight of 68 g/m²) size, and Canon PBPAPER having the A5 (148.5 mm in width×210 mm in length×83 μm in thickness, a basis weight of 64 g/m²) size were used. Additionally, in a case where the temperature of the film 51 in the non-sheet feeding area through which the papers P do not pass at the time of sheet feeding is measured, and the temperature exceeds 200° C., the interval (paper interval) between the preceding paper and the subsequent paper is increased. The maximum productivity refers to the productivity at the time when the temperature of the film 51 becomes 200° C. or less.

Embodiment 1 includes the heat generation members 54b3 and 54b4 for a plurality of small sizes corresponding to the B5 and A5 papers, and the temperature rise of the film 51 is small for any of the papers P, and the adjustment of the paper interval is not required. In Embodiment 1, the maximum productivity for the B5 paper was 39 sheets/minute, and the maximum productivity for the A5 paper was 46 sheets/minute. On the other hand, in Comparison Example 1, since only one kind of heat generation member 202 corresponding to the B5 paper is provided as the heat generation member, when printing B5 papers, the adjustment of the paper interval was not required, and the maximum productivity was 39 sheets/minute. However, since the heat generation member 202 corresponding to the B5 paper is used even when printing A5 papers, the temperature rise of the film 51 was large, and it was necessary to increase the paper interval so that the temperature rise in the non-sheet feeding portion will not occur, and it was found that the maximum productivity was as low as 16 sheets/minute.

As described above, according to Embodiment 1, since the heat generation member having a first length includes two heat generation members, i.e., a first heat generation member and a second heat generation member, the power provided to the heat generation member having the first length can be distributed. Additionally, since the power is always supplied to the first heat generation member and the second heat generation member at the same time, the temperature rise does not unevenly occur only in one end of the substrate in the width direction. Accordingly, assuming an unexpected apparatus failure, even if an electric power is excessively supplied to the heat generation member having the first length, the temperature gradient generated in the substrate in the width direction can be reduced. The fact that the temperature gradient is small enables the reduction of

distortion (heat stress) generated in the substrate, and the deformation of the substrate can be suppressed.

Next, the power supply capability of a third heat generation member and a fourth heat generation member having the lengths shorter than the first length in the longitudinal direction, and having different lengths in the longitudinal direction is made smaller than that of the heat generation member having the first length. Then, the third heat generation member and the fourth heat generation member are arranged between the first heat generation member and the second heat generation member in the width direction of the substrate, and the symmetry in the width direction of the substrate is maintained as much as possible. Accordingly, assuming an unexpected apparatus failure, even if an electric power is excessively supplied to one of the third heat generation member and the fourth heat generation member, the temperature gradient generated in the substrate in the width direction can be reduced, and the deformation of the substrate due to distortion can be suppressed. Then, since the third heat generation member and fourth heat generation member having the lengths shorter than the first length in the longitudinal direction, and having different lengths in the longitudinal direction are provided, the productivity for a plurality of kinds of papers having narrow widths can be improved. Finally, the reduction of the sizes of the heater can also be achieved at the same time by including two heat generation members only for the heat generation members having the first length, and including one heat generation member for each of the other heat generation members having shorter lengths in the longitudinal direction.

Modification 1

In Embodiment 1, although the details have been described about the configuration in which the two longest heat generation members 54b1 and 54b2 are electrically connected in parallel, and the power is supplied to the two longest heat generation members 54b1 and 54b2 at the same time, the configuration is not limited to this configuration. FIG. 9A is a diagram illustrating the configuration of the heater 54, and FIG. 9B is a diagram illustrating the heater 54 and the power control unit 97. As illustrated in FIG. 9A, the heater may be a heater in which the first contact 54d1, the first heat generation member 54b1, the second heat generation member 54b2, and the second contact 54d3 are electrically connected in series in this order. Specifically, in the heat generation member 54b1, one end is connected to the contact 54d1, and the other end is connected to the other end of the heat generation member 54b2 via the conductor 54c without any contacts. In the heat generation member 54b2, one end is connected to the contact 54d3, and the other end is connected to the other end of the heat generation member 54b1 via the conductor 54c without any contacts. In the heat generation member 54b3, one end is connected to the contact 54d1, and the other end is connected to the contact 54d3. In the heat generation member 54b4, one end is connected to the contact 54d3, and the other end is connected to the contact 54d4.

As illustrated in FIG. 9B, the electromagnetic relay 57a includes the contact 57a1 connected to the first pole of the AC power supply 55 via the triac 56, the contact 57a2 connected to the contact 54d1, and the contact 57a3 connected to the contact 54d4. The electromagnetic relay 57a is brought into either one of the states, i.e., the state where the contact 57a1 and the contact 57a2 are connected to each other, and the state where the contact 57a1 and the contact 57a3 are connected to each other, by the control of the

engine controller 92. The electromagnetic relay 57b includes the contact 57b1 connected to the second pole of the AC power supply 55, the contact 57b2 connected to the contact 54d2, and the contact 57b3 connected to the contact 54d3. The electromagnetic relay 57b is brought into either one of the states, i.e., the state where the contact 57b1 and the contact 57b2 are connected to each other, and the state where the contact 57b1 and the contact 57b3 are connected to each other, by the control of the engine controller 92.

FIG. 9A illustrates the electromagnetic relays 57a and 57b at the time of non-operation, the contact 57a1 and the contact 57a2 are connected to each other in the electromagnetic relay 57a, and the contact 57b1 and the contact 57b2 are connected to each other in the electromagnetic relay 57b. At the time of non-operation of the electromagnetic relays 57a and 57b, since power is supplied between the contact 54d1 and the contact 54d2, the longest heat generation members 54b1 and 54b2 generate heat.

In a case where only the electromagnetic relay 57b is operated, the contact 57a1 and the contact 57a2 are connected to each other in the electromagnetic relay 57a, and the electromagnetic relay 57b is brought into the state where the contact 57b1 and the contact 57b3 are connected to each other. At the time of operation of only the electromagnetic relay 57b, since power is supplied between the contact 54d1 and the contact 54d3, only the heat generation member 54b3 generates heat. In a case where only the electromagnetic relay 57a is operated, the contact 57a1 and the contact 57a3 are connected to each other in the electromagnetic relay 57a, and the electromagnetic relay 57b is brought into the state where the contact 57b1 and the contact 57b2 are connected to each other. At the time of operation of only the electromagnetic relay 57a, since power is supplied between the contact 54d4 and the contact 54d2, only the heat generation member 54b4 generates heat.

As described above, in FIG. 9A and FIG. 9B of the modification, one ends of the heat generation member 54b1 and the heat generation member 54b3 are electrically connected to the contact 54d1, which is the first contact. One ends of the heat generation member 54b4 and the heat generation member 54b2 are electrically connected to the contact 54d2, which is the second contact. The other end of the heat generation member 54b3 is electrically connected to the contact 54d3, which is the third contact. The other end of the heat generation member 54b4 is electrically connected to the contact 54d4, which is the fourth contact. Then, the other end of the heat generation member 54b1 and the other end of the heat generation member 54b2 are electrically connected to each other.

Also in the configuration of FIG. 9A and FIG. 9B, since it is the configuration in which power is supplied to the longest heat generation members 54b1 and 54b2 at the same time, the same effects as those in Embodiment 1 are exhibited. The supplyable power to the longest heat generation members 54b1 and 54b2 can be made equivalent to that in Embodiment 1, and the electrical resistance across both ends of each of the first heat generation member 54b1 and the second heat generation member 54b2, which are the longest heat generation members, may be 552. In FIG. 9A and FIG. 9B, the heat generation member 54b1 and the heat generation member 54b2 are connected in series, and the combined resistance value is 10 Ω . The other heat generation members may be the same as those in Embodiment 1. In this manner, also in Modification 1, the combined resistance of the heat generation member 54b1 and the heat generation member 54b2 is 10 Ω , and is smaller than the resistances (30 Ω) of the heat generation member 54b3 and the heat generation mem-

ber 54b4. The effects exhibited by the heater 54 illustrated in FIG. 9A and FIG. 9B are the same as those in Embodiment 1.

Modification 2

In Embodiment 1, although the details have been described about the case where the number of the non-longest heat generation members 54b3 and 54b4 are two, the configuration is not limited to this configuration. For example, as illustrated in FIG. 10, even with the configuration in which the number of the non-longest heat generation members is three, the same effects described in Embodiment 1 can be exhibited. That is, Modification 2 includes a heat generation member 54b5, which is a fifth heat generation member whose length in the longitudinal direction is shorter than that of the heat generation member 54b4, which is the fourth heat generation member. In the heat generation member 54b1 and the heat generation member 54b2, one ends are connected to the contact 54d1, which is a first common contact, and the other ends are connected to the contact 54d2, which is a second common contact. In the heat generation member 54b3, one end is connected to the contact 54d3, which is the third contact, and the other end is connected to the contact 54d2. In the heat generation member 54b4, one end is connected to the contact 54d4, which is the fourth contact, and the other end is connected to the contact 54d2. In the heat generation member 54b5, one end is connected to the contact 54d5, which is a fifth contact, and the other end is connected to the contact 54d2. That is, the other ends of all the heat generation members 54b1 to 54b5 are connected to the contact 54d2. Additionally, the three heat generation members 54b3 to 54b5 are arranged between the two heat generation members 54b1 and 54b2 in the width direction of the substrate 54a. Further, the heat generation member 54b5 is arranged between the heat generation members 54b3 and 54b4 in the width direction of the substrate 54a.

The heater 54 illustrated in FIG. 10 will be described. The longest heat generation members 54b1 and 54b2 are arranged on the both ends of the substrate 54a in the width direction, and power is supplied from the common contacts 54d1 and 54d2 to the longest heat generation members 54b1 and 54b2 at the same time. As in Embodiment 1, the electrical resistance across both ends of each of the longest heat generation members 54b1 and 54b2 is set to 20[Ω]. The lengths of the heat generation members 54b1 and 54b2 in the longitudinal direction are 222 mm.

The lengths in the longitudinal direction are 188 mm in the heat generation member 54b3, 154 mm in the heat generation member 54b4, and 111 mm in the heat generation member 54b5. The heat generation member 54b3 is used at the time of printing of a B5 paper, the heat generation member 54b4 is used for printing of an A5 paper, and the heat generation member 54b5 is used at the time of printing of an A6 paper. The electrical resistance across both ends of each of these non-longest heat generation members 54b3 to 54b5 is set to 30[Ω]. In this manner, also in Modification 2, the combined resistance of the heat generation member 54b1 and the heat generation member 54b2 is 10 Ω , and is smaller than the resistances (30 Ω) of the heat generation member 54b3 to the heat generation member 54b5. By increasing the number of kinds of the non-longest heat generation members to three, the maximization of the productivity for the three kinds of papers, a B5 paper, an A5 paper and an A6 paper, is enabled.

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In the non-longest heat generation members, assuming an excessive electric power supply, the power supplied to each of the heat generation members **54b3** to **54b5** is the same. Since the length of the heat generation member **54b5** in the longitudinal direction is the shortest, the degree of concentration of power is the highest, and the deformation risk of the substrate **54a** at the time of temperature rise is high. For the purpose of removing this risk as much as possible, the shortest heat generation member **54b5** can be arranged in the center portion in the width direction of the substrate **54a** to give the symmetry in the width direction. Additionally, the heat generation members **54b3** and **54b4** can be arranged on both sides of the heat generation member **54b5** in the width direction, to be close to the center as much as possible. The effects exhibited by the heater **54** illustrated in FIG. 10 are the same as those in Embodiment 1.

Modification 3

In Modification 2, four contacts are arranged at one end of the substrate **54a** in the longitudinal direction, and one contact is arranged at the other end. In Modification 3, an example will be described in which three contacts are arranged at one end in the longitudinal direction, and two contacts are arranged at the other end. In Modification 3, since the heat generation member can be arranged in the center in the longitudinal direction of the substrate **54a** to the utmost, it is an arrangement preferable for making the heat generation distribution in the longitudinal direction uniform.

Modification 3 includes the heat generation member **54b5**, which is the fifth heat generation member whose length in the longitudinal direction is shorter than that of the heat generation member **54b4**, which is the fourth heat generation member. In the heat generation member **54b1** and the heat generation member **54b2**, one ends are connected to the contact **54d1**, which is the first common contact, and the other ends are connected to the contact **54d2**, which is the second common contact. In the heat generation member **54b3**, one end is connected to the contact **54d3**, which is the third contact, and the other end is connected to the contact **54d2**. In the heat generation member **54b4**, one end is connected to the contact **54d3**, and the other end is connected to the contact **54d4**, which is the fourth contact. In the heat generation member **54b5**, one end is connected to the contact **54d5**, which is the fifth contact, and the other end is connected to the contact **54d4**. Among the five heat generation members, the first heat generation member **54b1** and the second heat generation member **54b2** having the longest length, and the fourth heat generation member **54b3** having the second longest length are connected to the second contact **54d2**. The fourth heat generation member **54b3** having the second longest length, and the fourth heat generation member **54b4** having the third longest length are connected to the third contact **54d3**. The fourth heat generation member **54b4** having the third longest length, and the fifth heat generation member **54b5** having the fourth longest length are connected to the fourth contact **54d4**. That is, the heat generation member **54b** is connected to the contact common to another heat generation member **54b** with which the difference in length from the heat generation member **54b** is the minimum. Additionally, the three heat generation members **54b3** to **54b5** are arranged between the two heat generation members **54b1** and **54b2** in the width direction of the substrate **54a**. Further, the heat generation member **54b5** is arranged between the heat generation members **54b3** and **54b4** in the width direction of the substrate **54a**.

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The heater **54** illustrated in FIG. 11 will be described. The longest heat generation members **54b1** and **54b2** are arranged on the both ends of the substrate **54a** in the width direction, and power is supplied from the common contacts **54d1** and **54d2** to the longest heat generation members **54b1** and **54b2** at the same time. As in Embodiment 1, the electrical resistance across both ends of each of the longest heat generation members **54b1** and **54b2** is set to $20[\Omega]$. The lengths of the heat generation members **54b1** and **54b2** in the longitudinal direction are 222 mm.

The lengths in the longitudinal direction are 188 mm in the heat generation member **54b3**, 154 mm in the heat generation member **54b4**, and 111 mm in the heat generation member **54b5**. The heat generation member **54b3** is used at the time of printing of a B5 paper, the heat generation member **54b4** is used for printing of an A5 paper, and the heat generation member **54b5** is used at the time of printing of an A6 paper. The electrical resistance across both ends of each of these non-longest heat generation members **54b3** to **54b5** in the longitudinal direction is set to $30[\Omega]$. In this manner, also in Modification 3, the combined resistance of the heat generation member **54b1** and the heat generation member **54b2** is 10Ω , and is smaller than the resistances (30Ω) of the heat generation member **54b3** to the heat generation member **54b5**. By increasing the number of kinds of the non-longest heat generation members to three, the maximization of the productivity for the three kinds of papers, a B5 paper, an A5 paper and an A6 paper, is enabled.

Assuming an excessive electric power supply in the non-longest heat generation members **54b**, the power supplied to each of the heat generation members **54b3** to **54b5** is the same. Since the length of the heat generation member **54b5** in the longitudinal direction is the shortest, the degree of concentration of power is the highest, and the deformation risk of the substrate **54a** at the time of temperature rise is high. For the purpose of removing this risk as much as possible, the shortest heat generation member **54b5** can be arranged in the center portion in the width direction of the substrate **54a** to give the symmetry in the width direction. Additionally, the heat generation members **54b3** and **54b4** can be arranged on both sides of the heat generation member **54b5** in the width direction, to be close to the center as much as possible. The effects exhibited by the heater **54** illustrated in FIG. 11 are the same as those in Embodiment 1.

Conventionally, the resistance of each of a plurality of heat generation members has the same resistance value, and the suppliable power is also the same. Conventionally, in a case where power is continuously supplied to a heat generation member having a wide width, an excessive temperature rise occurs in one end of a substrate in the width direction. Therefore, the temperature gradient in the substrate becomes large, and there is a possibility that the substrate is greatly distorted. Additionally, conventionally, since only one kind of a heat generation member having a narrow width is provided, in papers having a plurality of kinds of sizes, it is difficult to suppress the temperature rise in the non-sheet feeding area, and it is difficult to provide a high productivity. On the other hand, according to Embodiment 1, the deformation of a substrate on which a heater is mounted can be suppressed.

Embodiment 2

Since the shape of the heater **54** of Embodiment 2 is the same as that in Embodiment 1, and is as illustrated in FIG. 4, a description will be omitted. In Embodiment 2, among the non-longest heat generation members **54b3** and **54b4**,

the power density (described later) of the shorter heat generation member **54b4** is made higher than the power density of the longer heat generation member **54b3**. The non-longest heat generation members **54b3** and **54b4** have a large non-heating area that cannot be heated in the longitudinal direction. The shorter the length in the longitudinal direction of the heat generation member **54b** is, the wider this non-heating area becomes, and the heat of the heat generation member **54b** is easily taken away by the non-heating area. The fixing apparatus **50** cannot sufficiently perform heating in the vicinity of this non-heating area, and there is a possibility that a toner image cannot be fixed to the paper P. Therefore, at least the power density of the shorter heat generation member **54b4** can be made higher than the power density of the longer heat generation member **54b3**.

Additionally, among the non-longest heat generation members **54b3** and **54b4**, the resistance value of the shorter heat generation member **54b4** is made to be equal to or higher than the resistance value of the longer heat generation member **54b3**. Accordingly, the fixing apparatus **50** can be operated with a certain current amount or less, irrespective of whether the shorter heat generation member **54b4** or the longer heat generation member **54b3** is used. Accordingly, low rating and low cost wires, elements, etc. can be chosen for bundled wires, electric elements, etc. to be connected to the non-longest heat generation members **54b3** and **54b4**.

Here, the power density is defined as the value (in the unit of W/mm) obtained by dividing the power generated when 100V is provided to the heat generation member **54b** by the length of the heat generation member **54b** in the longitudinal direction. Let the electric resistance value of the longer heat generation member **54b3** be R1, the electric resistance value of the shorter heat generation member **54b4** be R2, the length of the longer heat generation member **54b3** in the longitudinal direction be L1, and the length of the shorter heat generation member **54b4** in the longitudinal direction be L2. In that case, the power of the longer heat generation member **54b3** is expressed by " $100^2/R1$ ", and the power of the shorter heat generation member **54b4** is expressed by " $100^2/R2$." Since the respective powers are divided by the length of the heat generation member **54b**, the power density of the longer heat generation member **54b3** is expressed by " $100^2/R1/L1$ ", and the power density of the shorter heat generation member **54b4** is expressed by " $100^2/R2/L2$." Embodiment 2 has the characteristic in the relationship " $100^2/R1/L1 < 100^2/R2/L2$." This relational expression can also be expressed as " $R1L1 > R2L2$."

[Power Density and Whether or not Fixing can be Performed]

The power density of the heat generation member **54b**, and the confirmation conditions for confirming whether fixing of a toner image to the paper P can be performed will be described below. The image process speed of an image forming apparatus is 200 mm/sec, and the interval (paper interval) between the preceding paper and the subsequent paper is set to 0.25 second. Sheet feeding is performed by performing the temperature control by the engine controller **92**, so that the back surface of the substrate **54a** becomes 180° C. by the fixing temperature sensor **59** installed in the back surface of the substrate **54a**. Note that the fixing apparatus **50** including the heater **54** is kept in the state where it is sufficiently cooled.

Among the non-longest heat generation members **54b3** and **54b4**, when using the longer heat generation member **54b3**, Canon CS680 paper having the B5 (182 mm in width×257 mm in length×92 μm in thickness, a basis weight of 68 g/m²) size is used. When using the shorter heat

generation member **54b4**, the above-described CS680 paper is cut into the A5 size (148.5 mm in width×210 mm in length×92 μm in thickness, a basis weight of 68 g/m²), and feeding of 10 papers are continuously performed in any case. Note that the toner image on the paper P is uniformly formed in the entire area of the paper P (each of the top margin, the bottom margin, the left margin, and the right margin is set to 5 mm), and a toner amount is 1.0 mg/cm².

Whether or not there is a portion in which the toner image on the paper P is unfixed is confirmed, and the case where all is fixed is considered to have no fixability problem and indicated by "○", and the case where there is an unfixed portion is considered to have a fixation failure and indicated by "x". The fixability is confirmed for the five kinds of longer heat generation members **54b3** having different power densities, and for the five kinds of shorter heat generation members **54b4** having different power densities. The confirmation results are illustrated in Table 1.

TABLE 1

heat generation member length	power density	fixability
longer heat generation member		
188	1.90	pass
188	1.77	pass
188	1.72	pass
188	1.66	fail
188	1.56	fail
shorter heat generation member		
154	2.03	pass
154	1.91	pass
154	1.80	pass
154	1.76	fail
154	1.71	fail

In Table 1, the left side table illustrates the longer heat generation member **54b3**, and the right side table illustrates the shorter heat generation member **54b4**. In each table, the length of the heat generation member **54b** in the longitudinal direction is shown in the first row, the power density is shown in the second row, and the above-described fixability (○ or x) is shown in the third row.

As illustrated in Table 1, in the longer heat generation member **54b3**, the entire toner image was fixed to the paper P with the power density of 1.72 [W/mm] or more, and there was no problem in the fixability. Additionally, in the shorter heat generation member **54b4**, the entire toner image was fixed to the paper P with the power density of 1.8 [W/mm] or more, and there was no fixability problem. Further, it was able to confirm that the heat generation member **54b4**, having a larger non-heating area in which heat is easily taken away by the non-heating area near the ends of the heat generation member **54b4**, and having a shorter length in the longitudinal direction, required a higher power density compared with the heat generation member **54b3**.

[Maximum Current Amount and Whether or not Fixing can be Performed]

Here, the maximum current amount refers to the current amount that flows when 100V is applied to the heat generation member **54b**. The smaller the value of this maximum current amount is, the more it is enabled to choose low cost and low rating wires, elements, etc. for bundled wires, electric elements, etc. to be connected to the heat generation member **54b**. FIG. 12 illustrates the relationship between the

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maximum current amount [A] and the power density [W/mm], and indicates the cases without a fixability problem with “○”, and the cases with a fixation failure with “x”.

In the longer heat generation member **54b3**, it is a plot Lg1 that has “○” for the fixability, and has the smallest maximum current amount. In the plot Lg1, the power density is 1.72 [W/mm], and the maximum current amount is 3.23 [A]. The electrical resistance of the heat generation member **54b3** at this time is 31[Ω]. In the shorter heat generation member **54b4**, it is a plot St1 that has “○” for the fixability, and has the smallest maximum current amount. In the plot St1, the power density is 1.80 [W/mm], and the maximum current amount is 2.78 [A]. The electrical resistance of the heat generation member **54b4** at this time is 36[Ω]. That is, in the shorter heat generation member **54b4** of the plot St1, the power density becomes higher, and the resistance value also becomes higher compared with the longer heat generation member **54b3** of the plot Lg1. In this manner, assuming that the longer heat generation member **54b3** is 31[Ω], and the shorter heat generation member **54b4** is 36[Ω], the fixability can be satisfied, and the maximum current amount can be kept to 3.23 [A] or less. Then, low cost and low rating wires, elements, etc. can be chosen for bundled wires, electric elements, etc. to be connected to the heat generation member **54b**.

Note that, in the shorter heat generation member **54b4**, although the conditions of the plot St1 were recommended, also in a plot St2 indicated by a black dot, the power density is as low as 2.09 [W/mm], and the maximum current amount is 3.23 [A] or less. The electric resistance value of the shorter heat generation member **54b4** at this time is 31[Ω]. Even if the electrical resistances are set to the same value, i.e., 31[Ω] for the longer heat generation member **54b3**, and 31[Ω] for the shorter heat generation member **54b4**, the fixability can be satisfied, and the maximum current amount can be kept to 3.23 [A] or less. That is, in the shorter heat generation member **54b4** of the plot St2, the power density becomes higher, and the resistance value is equal compared with the longer heat generation member **54b3** of the plot Lg1. From the above, in the graph of FIG. 12, the shorter heat generation member **54b4** can be used in the range from the plot St1 to the plot St2.

From the above confirmation results, among the non-longest heat generation members **54b3** and **54b4**, the power density of the shorter heat generation member **54b4** is made higher than the power density of the longer heat generation member **54b3**. Accordingly, irrespective of which one of the heat generation members **54b** is used, the fixability near the non-heating area in the both sides of the heat generation member **54b** can be satisfied. Further, by making the resistance value of the shorter heat generation member **54b4** equal to or higher than the resistance value of the longer heat generation member **54b3**, the fixing apparatus **50** can be operated with a certain current amount or less, and inexpensive bundled wires, etc. can be used.

As described above, according to Embodiment 2, the deformation of the substrate on which the heater is mounted can be suppressed.

Embodiment 3

FIG. 13A is a cross-sectional view of a fixation nip portion N of the fixing apparatus **50**, and illustrates a part of the film **51**, a part of the nip forming member **52**, the heater **54** and the pressure roller **53**. It is assumed that the center of the rotation axis of the pressure roller **53** is C, among the non-longest heat generation members **54b3** and **54b4**, the

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position of the shorter heat generation member **54b4** is H1, and the position of the longer heat generation member **54b3** is H2. The distance from the center C to the position H1 is defined as RL1, and the distance from the center C to the position H2 is defined as RL2. Embodiment 3 is characterized in that the heater **54** is arranged at a position where the distance RL1 becomes smaller than the distance RL2 (RL1 < RL2). Since the closer the distance between the center C of the pressure roller **53** and the heat generation member **54b** is, the greater the amount of collapse of the elastic layer of the pressure roller **53** becomes, the pressure in the fixation nip portion N at the position H1 can be made higher than that at the position H2.

FIG. 13B illustrates the profile of the pressure (nip pressure) of the fixation nip portion N in the conveyance direction of the paper P. In FIG. 13B, the horizontal axis represents the position in the conveyance direction corresponding to the fixation nip portion N illustrated in FIG. 13A, and the vertical axis represents the nip pressure. As illustrated in FIG. 13B, in the conveyance direction of the paper P, the nip pressure is the highest at the position of the center C of the pressure roller **53**. Additionally, as illustrated in FIG. 13B, it can be seen that the nip pressure at the position H1 is higher than the nip pressure at the position H2.

As described above, the distance from the position of the center of rotation of the pressure roller **53** to the heat generation member **54b** (the heat generation member **54b4** in FIG. 4, etc., and the heat generation member **54b5** in FIG. 10) having the shortest length in the longitudinal direction among the third heat generation member and the fourth heat generation member **54b** is RL1. The distance from the position of the center of rotation of the pressure roller **53** to the other heat generation members, except for the shortest heat generation member among the third heat generation member and the fourth heat generation member, is RL2. Then, in Embodiment 3, the heat generation members **54b** are arranged on the substrate at predetermined positions (for example, a center portion) in the longitudinal direction, so that the distance RL1 becomes shorter than the distance RL2.

Since the nip pressure is high, the thermal resistance due to contact can be reduced between the heater **54** and the film **51**, and between the film **51** and the pressure roller **53**, and the heat transfer property between each component can be improved. With this improvement in the heat transfer property, even if power is excessively supplied to the heat generation member **54b** at the time of occurrence of an unexpected failure, the excessive heat generated by the heater **54** can be quickly conducted to the pressure roller **53** having a high thermal capacity, etc. That is, the deformation risk of the substrate **54a** can be reduced.

Since the shorter the length of the heat generation member **54b** in the longitudinal direction is, the larger the non-heating area becomes, and the more heat is taken away, the power density of the shorter heat generation member **54b4** can be made higher than the power density of the longer heat generation member **54b3**. On the other hand, the risk of deformation of the substrate **54a** at the time of failure is slightly high. In order to reduce this risk, the shorter heat generation member **54b4** can be arranged at the position H1 having a higher nip pressure. In Embodiment 3, even if power is excessively supplied to the shorter heat generation member **54b4**, the generated heat can be quickly transferred to the pressure roller **53**, etc., and the risk of deformation of the substrate **54a** can be reduced. As described above, when incorporating the heater **54** described in Embodiment 1 and

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Embodiment 2 into the fixing apparatus 50, among the non-longest heat generation members 54b3 and 54b4, the shorter heat generation member 54b4 is arranged closer to the center C of the pressure roller 53 than the longer heat generation member 54b3. Accordingly, the risk of deformation of the substrate 54a can be reduced.

As described above, according to Embodiment 3, the deformation of the substrate on which the heater is mounted can be suppressed.

According to the present invention, the deformation of the substrate on which the heater is mounted can be suppressed.

While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

This application claims the benefit of Japanese Patent Application No. 2019-006469, filed Jan. 18, 2019, which is hereby incorporated by reference herein in its entirety.

What is claimed is:

1. A heater comprising:

a substrate;

a first heat generation member having a length in a longitudinal direction;

a second heat generation member having a length in the longitudinal direction, the length being substantially equal to the length of the first heat generation member;

a third heat generation member having a length in the longitudinal direction, the length being shorter than the lengths of the first heat generation member and the second heat generation member;

a fourth heat generation member having a length in the longitudinal direction, the length being shorter than the length of the third heat generation member; and

wherein the first heat generation member, the second heat generation member, the third heat generation member, and the fourth heat generation member are arranged on the substrate,

wherein the first heat generation member is arranged at one end of the substrate in a width direction,

wherein the second heat generation member is arranged at another end of the substrate in the width direction, to be symmetrical with the first heat generation member,

wherein the third heat generation member and the fourth heat generation member are arranged between the first heat generation member and the second heat generation member in the width direction of the substrate,

wherein the first heat generation member, the third heat generation member, the fourth heat generation member and the second heat generation member are arranged in this order in the width direction,

wherein a relationship of $R1 < R2$ is satisfied, and

wherein a relationship of $R1 \times L1 > R2 \times L2$ is satisfied, where $L1$ is the length of the third heat generation member in the longitudinal direction, $R1$ is a value of a resistance of the third heat generation member, $L2$ is the length of the fourth heat generation member in the longitudinal direction, and $R2$ is a value of a resistance of the fourth heat generation member.

2. A heater according to claim 1, wherein a value of a combined resistance of the first heat generation member and the second heat generation member is smaller than a value of a resistance of the third heat generation member, and a value of a resistance of the fourth heat generation member, and

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wherein the first heat generation member and the second heat generation member are electrically connected in parallel.

3. A heater according to claim 1, wherein the third heat generation member and the fourth heat generation member are arranged to be symmetrical in the width direction of the substrate.

4. A heater according to claim 1, further comprising:

a first contact to which one end of the first heat generation member and one end of the second heat generation member are electrically connected;

a second contact to which another end of the first heat generation member and another end of the second heat generation member are electrically connected;

a third contact to which another end of the third heat generation member and one end of the fourth heat generation member are electrically connected; and

a fourth contact to which another end of the fourth heat generation member is electrically connected.

5. A heater according to claim 1, further comprising:

a first contact to which one end of the first heat generation member and one end of the third heat generation member are electrically connected;

a second contact to which one end of the fourth heat generation member and one end of the second heat generation member are electrically connected;

a third contact to which another end of the third heat generation member is electrically connected; and

a fourth contact to which another end of the fourth heat generation member is electrically connected, wherein another end of the first heat generation member and another end of the second heat generation member are electrically connected to each other.

6. A heater according to claim 1, further comprising:

a first contact to which one end of the first heat generation member and one end of the second heat generation member are electrically connected;

a second contact to which another end of the first heat generation member and another end of the second heat generation member are electrically connected;

a third contact to which another end of the third heat generation member and one end of the fourth heat generation member are electrically connected; and

a fourth contact to which another end of the fourth heat generation member is electrically connected.

7. A fixing apparatus for fixing an unfixed toner image carried by a recording material, the fixing apparatus comprising:

a heater;

a first rotary member heated by the heater; and

a second rotary member forming a nip portion with the first rotary member,

wherein the heater comprising:

a substrate;

a first heat generation member having a length in a longitudinal direction;

a second heat generation member having a length in the longitudinal direction, the length being substantially equal in to a length of the first heat generation member;

a third heat generation member having a length in the longitudinal direction, the length being shorter than the lengths of the first heat generation member and the second heat generation member;

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a fourth heat generation member having a length in the longitudinal direction, the length being shorter than the length of the third heat generation member; and wherein the first heat generation member, the second heat generation member, the third heat generation member, and the fourth heat generation member are arranged on the substrate, wherein the first heat generation member is arranged at one end of the substrate in a width direction, wherein the second heat generation member is arranged at another end of the substrate in the width direction, to be symmetrical with the first heat generation member, wherein the third heat generation member and the fourth heat generation member are arranged between the first heat generation member and the second heat generation member in the width direction of the substrate, wherein the first heat generation member, the third heat generation member, the fourth heat generation member and the second heat generation member are arranged in this order in the width direction, wherein a relationship of $R1 < R2$ is satisfied, and wherein a relationship of $R1 \times L1 > R2 \times L2$ is satisfied, where $L1$ is the length of the third heat generation member in the longitudinal direction, $R1$ is a value of a resistance of the third heat generation member, $L2$ is the length of the fourth heat generation member in the longitudinal direction, and $R2$ is a value of a resistance of the fourth heat generation member.

8. A fixing apparatus according to claim 7, wherein the first rotary member is a film.

9. A fixing apparatus according to claim 8, wherein the heater is provided in an inner space of the film, wherein the heater and the second rotary member pinch the film, and wherein an image on a recording material is heated at the nip portion formed between the film and the second rotary member, through the film.

10. A fixing apparatus according to claim 7, wherein at a predetermined position in the longitudinal direction, a distance from a position of a center of rotation of the second rotary member to a heat generation member having a shortest length in the longitudinal direction among other heat generation members except for the first heat generation member and the second heat generation member is shorter than a distance from the position of the center of rotation of the second rotary member to a heat generation member except for the heat generation member having the shortest length among the other heat generation members.

11. An image forming apparatus comprising:
an image forming unit configured to form an unfixed toner image on a recording material; and
a fixing apparatus according to claim 7,
wherein the fixing apparatus is configured to fix the unfixed toner image on the recording material.

12. An image forming apparatus according to claim 11, wherein a value of a combined resistance of the first heat generation member and the second heat generation member

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is smaller than a value of a resistance of the third heat generation member, and a value of a resistance of the fourth heat generation member.

13. An image forming apparatus according to claim 11, wherein the third heat generation member and the fourth heat generation member are arranged to be symmetrical in the width direction of the substrate.

14. An image forming apparatus according to claim 11, further comprising:

a first contact to which one end of the first heat generation member and one end of the second heat generation member are electrically connected;

a second contact to which another end of the first heat generation member and another end of the second heat generation member are electrically connected;

a third contact to which another end of the third heat generation member and one end of the fourth heat generation member are electrically connected; and

a fourth contact to which another end of the fourth heat generation member is electrically connected.

15. An image forming apparatus according to claim 11, further comprising:

a first contact to which one end of the first heat generation member and one end of the third heat generation member are electrically connected;

a second contact to which one end of the fourth heat generation member and one end of the second heat generation member are electrically connected;

a third contact to which another end of the third heat generation member is electrically connected; and

a fourth contact to which another end of the fourth heat generation member is electrically connected,

wherein another end of the first heat generation member and another end of the second heat generation member are electrically connected to each other.

16. A fixing apparatus according to claim 7, wherein a value of a combined resistance of the first heat generation member and the second heat generation member is smaller than a value of a resistance of the third heat generation member, and a value of a resistance of the fourth heat generation member.

17. A fixing apparatus according to claim 7, wherein the third heat generation member and the fourth heat generation member are arranged to be symmetrical in the width direction of the substrate.

18. A fixing apparatus according to claim 7, further comprising:

a first contact to which one end of the first heat generation member and one end of the third heat generation member are electrically connected;

a second contact to which one end of the fourth heat generation member and one end of the second heat generation member are electrically connected;

a third contact to which another end of the third heat generation member is electrically connected; and

a fourth contact to which another end of the fourth heat generation member is electrically connected,

wherein another end of the first heat generation member and another end of the second heat generation member are electrically connected to each other.

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